

## IMPLEMENTATION REPORT

# Centro

AI-powered document collection platform for accounting firms — Pilot MVP build

|                      |                                                                 |
|----------------------|-----------------------------------------------------------------|
| Prepared for         | noammaarachot@gmail.com                                         |
| Source spec          | Centro Engineering Product Specification (EPS) v2.0, Ch. 1–26   |
| Report date          | July 19, 2026                                                   |
| Milestones delivered | M1 – M13 (13 of 13), all committed to git                       |
| Commits              | 13 (1 pre-existing landing page + 12 build milestones)          |
| Lines changed        | +19,617 / –4,053 across 74 files                                |
| Stack                | Next.js 16 (App Router) · TypeScript · Drizzle ORM · PostgreSQL |

# 1. Executive Summary

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Centro's Pilot MVP was built from the EPS end to end across thirteen milestones (M1–M13), each independently verified with scripted, real HTTP end-to-end tests against a live embedded Postgres database — not just type-checks. Every EPS chapter with pilot-scope requirements (Ch.1–Ch.20, plus the operational chapters 21–26) has corresponding working code, with the exception of pieces that require external credentials this environment doesn't have (Google OAuth, WhatsApp Business API, an LLM/OCR provider) — those are implemented as clearly-documented mock adapters with identical interfaces, so a real integration is a contained swap, not a redesign.

The single largest event during the build was a security audit (part of M12) that found and fixed a real, severe cross-tenant data-isolation bug: several server actions mutated resources by ID alone, without confirming the resource belonged to the caller's organization. This is detailed in full in Section 8 — it was found, fixed, and independently re-verified with a scripted cross-organization attack before the milestone was considered complete.

|                         |                                                                                                                 |
|-------------------------|-----------------------------------------------------------------------------------------------------------------|
| Milestones              | 13 of 13 complete                                                                                               |
| Database tables         | 13 (7 migrations, 0000–0006)                                                                                    |
| App routes              | 18 pages/routes, 2 API endpoints                                                                                |
| Server actions          | ~45 across 9 action files                                                                                       |
| Scripted E2E checks run | 190+ individual assertions across 13 milestone test passes, all green at time of writing                        |
| Mocked integrations     | Google Drive, WhatsApp Business API, AI/OCR classification — real credentials not available in this environment |
| Real, non-mocked infra  | Auth/sessions, audit log, state machines, scheduler, business-hours gating, resilience/retry, security fixes    |

**How to read "mocked":** a mocked integration means the *transport* (the actual network call to Google/Meta/an LLM provider) is a deterministic stand-in, generating realistic IDs/responses instead of calling a real API. In every case, the *decision logic* around it — matching, validation, retries, gating, state transitions — is real, not a placeholder, and was built to the exact interface a real integration would need to implement, so swapping the transport later touches one file, not the surrounding system.

# 2. Table of Contents

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### 3. System Architecture

#### 3.1 Stack

| Layer        | Choice                                                  | Why                                                                                                                                                                                                                                  |
|--------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Framework    | Next.js 16.2.10 (App Router, Turbopack)                 | Already the project's stack; AGENTS.md flagged it as having breaking changes from training-data knowledge, so the actual v16 docs (bundled in <code>node_modules/next/dist/docs</code> ) were read before writing routing/data code. |
| Language     | TypeScript, strict mode                                 | Existing project convention.                                                                                                                                                                                                         |
| Database     | PostgreSQL via Drizzle ORM                              | Matches EPS DB-8.1 exactly ("PostgreSQL is the primary database").                                                                                                                                                                   |
| Local dev DB | PGLite (embedded WASM Postgres)                         | Zero-install developer experience — <code>npm install &amp;&amp; npm run dev</code> works with no local Postgres. Production requires a real <code>DATABASE_URL</code> (enforced at runtime).                                        |
| Auth         | Custom session-cookie auth, scrypt password hashing     | EPS Ch.13 specifies a single shared employee account per firm — no off-the-shelf multi-user auth library fit that model.                                                                                                             |
| Styling      | Tailwind CSS v4, existing design tokens                 | Reused the landing page's design system (Ch.26) for visual consistency.                                                                                                                                                              |
| Mutations    | React 19 Server Actions (progressive enhancement forms) | Idiomatic for Next.js 16; every mutation is a real HTTP-testable POST, which is how all E2E verification was done.                                                                                                                   |

#### 3.2 Module Map

Matches the EPS Ch.7 module table directly:

| EPS Module          | Implementation                                                                                                    |
|---------------------|-------------------------------------------------------------------------------------------------------------------|
| Authentication      | <code>src/lib/auth/</code> (session, password, rate limiter), <code>src/app/login/</code>                         |
| Client Management   | <code>src/app/(app)/clients/</code> , <code>src/app/(app)/services/</code>                                        |
| Collection Engine   | <code>src/lib/collectionRequestStateMachine.ts</code> , <code>src/app/(app)/collections/</code>                   |
| Messaging           | <code>src/lib/conversationOrchestration.ts</code> , <code>src/lib/scheduler.ts</code> (WhatsApp transport mocked) |
| Document Processing | <code>src/lib/ai/</code> (classification mocked, pipeline logic real)                                             |

|           |                                                                           |
|-----------|---------------------------------------------------------------------------|
| Storage   | <code>src/lib/storage/driveAdapter.ts</code> (Drive transport mocked)     |
| Dashboard | <code>src/app/(app)/dashboard/</code> , <code>src/app/(app)/audit/</code> |

### 3.3 Architectural Decisions Worth Knowing

- **Modular monolith** (AR-7.1) — one Next.js app, no microservices, matching pilot scope explicitly.
- **Mock-first integrations** — agreed with the user early: Google Drive, WhatsApp, and AI/OCR are all mocked behind adapter modules with the exact interface a real provider would need, because real credentials for any of them weren't available in this environment. See Section 9.
- **Single shared DB, org-scoped rows** — rather than one deployment per firm (which the EPS Ch.21 roadmap table literally shows for the pilot), every table carries `organization_id` and every query is scoped to it. This was an explicit decision the user confirmed, trading off against the "isolated deployment per firm" wording in favor of lower operational overhead for a pilot with few firms.
- **No public signup** — organizations are provisioned only via an internal CLI script ( `npm run org:create` ), never a public form. Also an explicit user decision, matching the EPS's sales-assisted tone rather than self-serve SaaS.
- **PGLite single-connection constraint** — discovered during testing: PGLite (used for local dev) does not reliably support two separate Node processes holding the same database directory open concurrently. All test mutations were routed through the running dev server's own HTTP endpoints rather than a parallel script once this was found.

## 4. Database & Data Model

13 tables across 7 migrations ( `drizzle/0000` – `0006` ), matching the EPS Ch.4/Ch.8 domain model plus the operational additions from later chapters (business-hours config, audit trail precision, Drive folder mapping).

| Table                                        | Purpose                                                                                                                                                                                                                                      | Added                |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <code>organizations</code>                   | Tenant root. Also carries onboarding/integration status (Ch.3) and Ch.18 scheduling config (business hours/days, reminder interval, inactivity timeout).                                                                                     | M1, extended M4 & M8 |
| <code>users</code>                           | The single shared employee account per organization (BR-13.1) — one row per org, not per person.                                                                                                                                             | M2                   |
| <code>sessions</code>                        | DB-backed session tokens (cookie-based auth).                                                                                                                                                                                                | M2                   |
| <code>audit_logs</code>                      | Immutable event log (Ch.17). Insert-only. Carries <code>organization_id</code> , <code>client_id</code> , <code>collection_request_id</code> (added M11 for FR-17.3 precision), actor type/user, event type, description, optional metadata. | M2, extended M11     |
| <code>clients</code>                         | Business records only, never authenticate (Ch.2). Unique ( <code>organization_id</code> , <code>phone</code> ) — one client per WhatsApp number. Carries <code>drive_folder_id</code> (M5).                                                  | M3, extended M4 & M5 |
| <code>services</code>                        | Recurring document-requirement definitions (BR-001: "Services define requirements; clients do not").                                                                                                                                         | M3                   |
| <code>service_document_requirements</code>   | Per-service requirement templates (e.g. "Bank Statement").                                                                                                                                                                                   | M3                   |
| <code>client_services</code>                 | Many-to-many Client↔Service assignment.                                                                                                                                                                                                      | M3                   |
| <code>collection_requests</code>             | One collection cycle for one client/service/period. Ch.6 state machine status enum.                                                                                                                                                          | M3                   |
| <code>collection_request_requirements</code> | Per-cycle <i>snapshot</i> of the service's requirement templates (BR-002: historical requests are immutable to later service edits).                                                                                                         | M3                   |
| <code>documents</code>                       | Metadata + Google Drive reference (never file bytes). Ch.6 status enum plus <code>deleted_from_drive</code> (Ch.14) with a <code>drive_deleted_at</code> timestamp.                                                                          | M3, extended M5      |
| <code>conversations</code>                   | One WhatsApp conversation per active Collection Request (BR-003). Ch.6 status enum including <code>human_control</code> .                                                                                                                    | M3                   |
| <code>messages</code>                        | Inbound/outbound message log, actor-typed sender.                                                                                                                                                                                            | M3                   |

Every table's primary key is a UUID (DB-8.2). Every business table carries `organization_id` (DB-8.4). Foreign keys use `onDelete: "cascade"` for true parent/child ownership (e.g. a deleted org cascades everything) but deliberately `onDelete: "restrict"` (the default) for business-history references — a Client or Service with real Collection Request history cannot be hard-deleted, surfaced as a friendly UI error rather than a crash. `audit_logs` is the one exception: its client/user references use `onDelete: "set null"`, since a deleted client must never become permanently undeletable-in-spirit just because it was once mentioned in a log entry (a real bug found and fixed in M3 — see Section 5).

## 5. Milestone-by-Milestone Build Log

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### M1 — Foundations & Environment

EPS Ch.7, Ch.19

Postgres/Drizzle data layer with automatic dev/prod driver selection ( `src/db/index.ts` ): production requires a real `DATABASE_URL` and refuses to start without one; local dev falls back to embedded PGlite with zero setup. Env var scaffolding ( `.env.example` ), migration tooling ( `npm run db:generate` / `db:migrate` ), and the `(app)` route group established as the internal-app boundary separate from the untouched marketing site.

**Bug found & fixed:** Turbopack's dev bundling broke PGlite's WASM/filesystem asset loading with an unrelated-looking "path argument" `TypeError` on every DB call — fixed via `serverExternalPackages` in `next.config.ts`. `next build`'s static analysis never exercises this path (DB routes are `force-dynamic` ), so it only surfaced under `next dev`.

### M2 — Auth, Organization & Audit Log

EPS Ch.2, Ch.13, Ch.17

Shared-account login (script password hashing, DB-backed sessions, `requireSession()` guarding every protected route), Organization + shared User creation (via the internal CLI provisioning script, per the no-public-signup decision), and `recordAuditEvent()` as the single insert-only write path onto the audit log.

Verified end-to-end over real HTTP: login reject/accept, protected-route redirect, logout, re-redirect after logout — including correctly decoding Next.js Server Actions' multipart wire format for scripted testing (not a trivial form POST).

### M3 — Core Domain Schema + Client/Service CRUD

EPS Ch.4, Ch.8

Full schema for every remaining entity. Hand-managed CRUD only for Clients and Services (the two entities an admin actually creates directly) — Collection

Requests/Documents/Conversations/Messages get their tables now but no "create" UI, since they're generated by the automation engine in later milestones, not typed in by hand.

**Bug found & fixed:** `audit_logs.client_id` had no `onDelete` behavior, so logging a client's own creation event (which happens immediately) made that client permanently undeletable forever after — the FK blocked it. Fixed with `onDelete: "set null"`: the log entry and its description text survive, only the structured reference clears.



## M4 — Onboarding

EPS Ch.3

Mocked Google/WhatsApp connection toggles, CSV client import (hand-written parser — see the Excel note in Section 11), onboarding summary, and the BR-001 automation-activation gate enforced server-side (not just by disabling a button). `clients.phone` made unique per organization, with friendly duplicate-handling in both the CSV import and the manual client form.

## M5 — Google Drive Storage (mocked)

EPS Ch.14, PR-005

`src/lib/storage/driveAdapter.ts` : `ensureClientFolder` (BR-3.003 — stores the folder ID, not its name, created lazily) and `uploadDocument`, wired to the real approval path (document approval in both the manual-add and review-document flows triggers upload the moment status becomes `approved`, per BR-11.5). Ch.14's manual-deletion reconciliation is fully implemented: a document "deleted from Drive" keeps its database row, flips status, and the UI shows "Deleted manually on DD/MM/YYYY HH:MM."

## M6 — Scheduler (the real automatic trigger)

EPS Ch.5, Ch.16, Ch.18

`src/lib/scheduler.ts` + `POST /api/cron/tick` (bearer-token gated via `CRON_SECRET`, required in production): finds conversations idle past the org's configured inactivity timeout and evaluates them; finds conversations stuck waiting for a client reply past the reminder interval and sends a reminder. This is the actual automatic trigger that M8's manual "run now" buttons had stood in for.

**Bug found & fixed:** the evaluation function updated the Conversation's status but never the Collection Request's own status (also part of the Ch.6 state machine) — meaning the scheduler's reminder query, which correctly checks the Collection Request status, could never have matched anything. Both now transition together.

## M7 — Collection Request Engine & State Machine

EPS Ch.5, Ch.6

Reordered ahead of M5/M6 mid-build (user-approved) once it became clear neither Drive nor WhatsApp integration had a real caller yet. The FR-6.2 transition graph ( `draft` → `active` → `waiting_for_client` → `processing` → `completed` / `escalated` / `cancelled` , with completion reachable only via `processing` , and reopening the one deliberate exception back to `active` ), plus the completion gate (BR-6.2: processing documents block completion; BR-11.2/BR-6.1: every requirement needs an approved document). Manual document marking (BR-11.3) lets an employee attach a document with a status directly — a real, spec-anticipated capability independent of the AI pipeline, not a placeholder for it.

## M8 — Conversation Orchestration

EPS Ch.10, Ch.16, Ch.18

`src/lib/conversationOrchestration.ts` is the WhatsApp transport boundary — one function (`sendOutboundMessage`) is the single call site a real Business API integration replaces. Real logic: `ensureConversation` (BR-003), the initial-request send, identical-filename duplicate detection (later superseded by M9's fuzzier version), `evaluateAndPrompt` (the "after N minutes of inactivity" decision, silent when unsatisfied), and `reopenIfCompleted` (a new document after completion reopens the request automatically). Since there was no scheduler yet at this point, the inactivity timer and inbound WhatsApp webhook were both stood in for by explicit buttons on the Collection Request page — M6 later supplied the real automatic trigger for the same functions.

## M9 — AI Document Intelligence (mocked)

EPS Ch.9, Ch.11

`src/lib/ai/intentClassifier.ts` and `documentClassifier.ts`: no OCR/LLM provider is configured, so both use deterministic rules (keyword matching for intent; filename-token overlap against candidate requirement names for document matching) behind the exact interface a real provider would implement. The full Ch.11 pipeline is wired for real: unsupported file types and unreadable files are rejected/flagged automatically before ever creating a document row; a confident match ( $\geq 60\%$ ) auto-approves and uploads to Drive; anything less certain — or unmatched entirely — is still filed but lands in `needs_review` for a person to resolve.

**Gap found & fixed:** a document classification couldn't confidently match had nowhere to render at all — the page only ever showed documents grouped under a requirement. Added an "Unmatched documents" section with a manual assign-to-requirement action.

## M10 — Operational Dashboard

EPS Ch.12

Five status cards (Active, Waiting for Client, Needs Employee Review, Documents in Processing, Completed Today) each with count and percentage of all active collections (FR-12.2). "Needs Employee Review" is a proper union (not a naive sum) of escalated requests and requests with a `needs_review` document, so a request matching both criteria only counts once. Drill-down per card (FR-12.3) shows client, progress %, missing documents, last activity, status — not a full client list. Client search by name or phone (FR-12.5).

## M11 — Audit Trail UI + Resilience

EPS Ch.15, Ch.17

The audit log itself existed since M2 but nothing ever displayed it — closed with an org-wide `/audit` feed and a per-Collection-Request "Audit History" section (FR-17.3), which required adding `collection_request_id` to the schema since `client_id` alone wasn't precise enough (one client can have many requests over time). `src/lib/resilience.ts` ( `withRetry` ): exponential-backoff retry wrapper wired into the Drive upload call site; callers catch failures and degrade gracefully (log + leave the document approved-but-unfiled) rather than crashing the action or corrupting the Collection Request (BR-15.1).

## M12 — Security Hardening

EPS Ch.13, NFR-24.4

See Section 8 for full detail. Found and fixed a severe cross-tenant IDOR affecting document approval and client/service CRUD; fixed a related authorization gap in the scheduler's manual trigger; added login rate limiting (5 attempts / 15 minutes) and standard security response headers.

## M13 — Full Pilot Acceptance Validation

EPS Ch.20

One continuous 32-assertion scripted run through the entire pilot lifecycle in a single organization — onboarding through business-hours config, service/requirement setup, CSV client import, collection request creation, conversation initiation, AI document classification (both auto-approved and needs-review paths), dashboard queue visibility, scheduler-driven prompting, client-confirmed completion, Drive deletion reconciliation, request reopening, and full audit trail verification. All 32 checks passed. Confirmed via `git diff` against the initial commit that the marketing landing page has zero changes across the entire build.

## 6. Routes, Pages & Server Actions Reference

### 6.1 Pages

| Route                                                                             | Purpose                                                                                                            | Auth                                             |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| <code>/</code>                                                                    | Marketing landing page (pre-existing, untouched)                                                                   | Public                                           |
| <code>/login</code>                                                               | Shared-account login                                                                                               | Public (redirects away if already authenticated) |
| <code>/dashboard</code>                                                           | Work-queue status cards, drill-down, client search                                                                 | Protected                                        |
| <code>/onboarding</code>                                                          | Integration toggles, CSV import, automation activation                                                             | Protected                                        |
| <code>/clients</code> , <code>/clients/new</code> , <code>/clients/[id]</code>    | Client CRUD, service assignment, collection-request creation                                                       | Protected                                        |
| <code>/services</code> , <code>/services/new</code> , <code>/services/[id]</code> | Service CRUD, requirement templates                                                                                | Protected                                        |
| <code>/collections</code> , <code>/collections/[id]</code>                        | Collection Request list/detail — status transitions, requirements, documents, conversation timeline, audit history | Protected                                        |
| <code>/audit</code>                                                               | Org-wide audit log                                                                                                 | Protected                                        |
| <code>/settings</code>                                                            | Business-hours/scheduling config, manual scheduler trigger                                                         | Protected                                        |

### 6.2 API Routes

| Route                       | Method | Purpose                                                                                                                                       |
|-----------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <code>/api/health</code>    | GET    | DB connectivity check (ops monitoring)                                                                                                        |
| <code>/api/cron/tick</code> | POST   | External-scheduler-callable endpoint running the real automatic evaluation/reminder trigger. Bearer-token gated by <code>CRON_SECRET</code> . |

### 6.3 Server Action Files

| File                                  | Actions                           |
|---------------------------------------|-----------------------------------|
| <code>src/app/login/actions.ts</code> | <code>login</code> (rate-limited) |
| <code>src/app/(app)/actions.ts</code> | <code>logout</code>               |

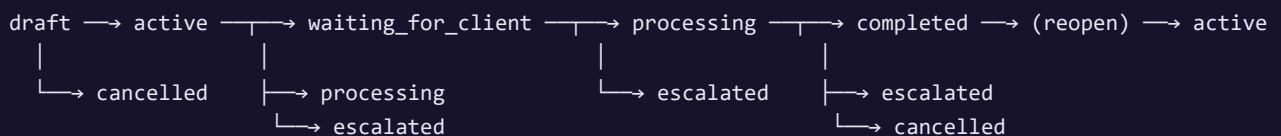
|                                                  |                                                                                                                                                                               |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| src/app/(app)/clients/actions.ts                 | createClient , updateClient ,<br>deleteClient , assignService ,<br>unassignService                                                                                            |
| src/app/(app)/services/actions.ts                | createService , updateService ,<br>deleteService , addRequirement ,<br>removeRequirement                                                                                      |
| src/app/(app)/collections/actions.ts             | createCollectionRequest ,<br>transitionStatus , addManualDocument ,<br>reviewDocument ,<br>assignDocumentRequirement ,<br>simulateDriveDeletion                               |
| src/app/(app)/collections/conversationActions.ts | initiateConversation ,<br>simulateInboundMessage , evaluateNow ,<br>markFinished , markMoreDocuments ,<br>takeOverConversation ,<br>releaseConversation , sendEmployeeMessage |
| src/app/(app)/onboarding/actions.ts              | connectGoogle / disconnectGoogle ,<br>connectWhatsapp / disconnectWhatsapp ,<br>activateAutomation , deactivateAutomation ,<br>importClients                                  |
| src/app/(app)/settings/actions.ts                | updateBusinessHours , runSchedulerNow                                                                                                                                         |

## 7. Core Workflows

### 7.1 Document Collection Lifecycle (Ch.5/Ch.6/Ch.10)

1. Employee assigns a Service to a Client, then starts a Collection Request for a period (or, once a scheduler-driven auto-create trigger is wired in a future milestone, this happens on its own).
2. The Service's requirement templates are snapshotted into the request (BR-002) — later edits to the Service template never retroactively change this request.
3. Employee (or, once real WhatsApp is connected, the automated flow) initiates the conversation — the first outbound message.
4. Documents arrive (simulated inbound messages today); each runs through the Ch.11 pipeline — validated, classified, matched, auto-approved or flagged for review, stored to (mock) Drive.
5. Once all requirements are satisfied, the scheduler (or an idle-triggered evaluation) prompts the client to confirm Finished / More Documents — never assumed automatically (BR-16.1).
6. A Finished reply walks the request through `processing → completed` ; a new document after completion reopens it automatically.

### 7.2 Collection Request State Machine



## 8. Security Review & the Cross-Tenant Fix

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### CRITICAL — FOUND AND FIXED

As part of M12, a dedicated audit was run specifically to check EPS NFR-24.4 ("each accounting firm shall operate in a fully isolated environment"): every database read/write reachable from a route param or bound server-action argument was checked for whether it verified the resource belonged to the caller's `organization_id`.

### What was found

Most mutations that took a resource ID as a bound server-action argument fetched or mutated the row by ID *alone*, without checking organization ownership:

- `clients/actions.ts` : `updateClient` , `deleteClient` , `assignService` , `unassignService`
- `services/actions.ts` : `updateService` , `deleteService` , `addRequirement` , `removeRequirement`
- `collections/actions.ts` : `reviewDocument` , `assignDocumentRequirement` , `simulateDriveDeletion` , `addManualDocument`

The concrete exploit: an authenticated employee of Organization A could view, edit, approve, reject, or delete Organization B's clients, services, and documents by substituting a foreign ID into an otherwise-legitimate request — the most severe form being direct cross-tenant approval/rejection of another firm's client documents.

A related, separate issue: the `/settings` "run scheduler now" button called the exact same function the global cron endpoint uses, so any employee's click would trigger message sends and evaluations across *every* organization in the database, not just their own.

### What was fixed

Every affected query now scopes explicitly to `organization_id = session.organizationId`, either directly or via shared helpers ( `getOrgScopedCollectionRequest` , `getScopedDocument` , `getOrgScopedService` ) that also verify parent-child ownership — e.g. a document must belong to the *specific* collection request it's addressed through, not just exist somewhere in the org. `createCollectionRequest` additionally verifies both the client and service belong to the caller's org before trusting a `client_services` assignment row. `runScheduledTasks` now takes an optional `organizationId` to scope a single run; only the cron endpoint omits it.

### How it was verified

1. Two real, separate organizations were provisioned. Direct-URL access from Org B to Org A's client/collection-request/service by ID correctly returned 404 in every case.
2. A genuine wire-level forgery: Org B's own legitimately-issued server-action reference (a real, validly-signed action ID) was submitted with the UUID *inside* the bound-argument payload swapped for Org A's client ID — not just a different URL, which Next.js Server Actions don't route by. The forged delete request returned normally (200/303) but affected zero rows; Org A's client was confirmed to still exist immediately after.

3. A full regression pass of every normal same-org CRUD/assignment/collection-request/document-approval flow (15 assertions) confirmed the added checks don't break legitimate access.

### Additional hardening in the same milestone

| Control             | Detail                                                                                                                                                                                                                                                                                 |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Login rate limiting | 5 failed attempts per 15-minute window locks out that email, even against a subsequently-correct password. Process-local (acceptable for a single-instance pilot per BR-13.2).                                                                                                         |
| Security headers    | <code>X-Frame-Options: DENY</code> , <code>X-Content-Type-Options: nosniff</code> , <code>Referrer-Policy: strict-origin-when-cross-origin</code> , <code>Permissions-Policy</code> , <code>Strict-Transport-Security</code> — set on every response via <code>next.config.ts</code> . |
| Secrets audit       | Confirmed no hardcoded credentials anywhere in source; only <code>.env.example</code> is tracked in git.                                                                                                                                                                               |
| Password storage    | Node's native <code>scrypt</code> (salted, timing-safe comparison) — no external dependency needed.                                                                                                                                                                                    |



## 9. Integrations — Mocked vs. Real

| Integration                                      | Status | Real interface implemented?                                                                                 | What's needed for real                                                                                                                                                                  |
|--------------------------------------------------|--------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Google Drive                                     | MOCKED | Yes — <code>ensureClientFolder</code> / <code>uploadDocument</code> in <code>driveAdapter.ts</code>         | Google Cloud project, OAuth client credentials, Drive API scope grant                                                                                                                   |
| WhatsApp Business API                            | MOCKED | Yes — <code>sendOutboundMessage</code> in <code>conversationOrchestration.ts</code> is the single call site | Meta Business verification (has its own real-world lead time), WhatsApp Business API access, webhook endpoint for inbound messages                                                      |
| AI/OCR document classification                   | MOCKED | Yes — <code>classifyDocument</code> / <code>classifyIntent</code> in <code>src/lib/ai/</code>               | An LLM API key (e.g. Anthropic) plus an OCR provider for image-based documents                                                                                                          |
| Authentication                                   | REAL   | —                                                                                                           | Nothing — fully implemented, not a stand-in                                                                                                                                             |
| Audit logging                                    | REAL   | —                                                                                                           | Nothing                                                                                                                                                                                 |
| Scheduler / cron trigger                         | REAL   | —                                                                                                           | An actual external cron caller (Vercel Cron, GitHub Actions scheduled workflow, etc.) needs to be pointed at <code>POST /api/cron/tick</code> — the endpoint itself is production-ready |
| State machines, business rules, resilience/retry | REAL   | —                                                                                                           | Nothing                                                                                                                                                                                 |

The mocking pattern is consistent throughout: each mock lives behind a small adapter module exposing the same function signatures a real integration would need, and every caller only ever touches the returned ID/result — never anything mock-specific. Swapping in a real provider is a change contained to that one file per integration.

## 10. Testing Performed

Every milestone was verified with a scripted, real end-to-end HTTP test — not just `npm run build` passing. Each test logs into a live dev server, drives the actual UI/forms via genuine POST requests (including correctly decoding Next.js Server Actions' multipart bound-argument wire format), and asserts on the resulting page content and/or direct database state. No test was declared passing without being run and observed green.

| Milestone | What was tested                                                                                                              | Assertions                           |
|-----------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| M1        | DB connectivity, migration tooling, build correctness                                                                        | Direct connection + query round-trip |
| M2        | Login/logout, session cookie behavior, protected-route redirects, audit trail accuracy                                       | Full HTTP flow, cookie inspection    |
| M3        | Client/Service CRUD, assignment, deletion blocking, audit FK fix                                                             | 11 checks                            |
| M4        | Onboarding gate, CSV import (valid/invalid/duplicate rows), duplicate-phone rejection                                        | 15 checks                            |
| M5        | Drive upload on approval, Drive-link display, deletion reconciliation                                                        | 8 checks                             |
| M6        | Scheduler evaluation branch, reminder branch, CRON_SECRET auth (401/200)                                                     | 9 checks + auth verification         |
| M7        | Full state machine walk, invalid-transition blocking, completion gate, reopen                                                | 18 checks                            |
| M8        | Conversation flow end-to-end, duplicate detection, human takeover/release                                                    | 25 checks                            |
| M9        | Unsupported/unreadable rejection, auto-approval, fuzzy duplicates, unmatched-document handling                               | 13 checks                            |
| M10       | Dashboard queue accuracy, search, queue transitions on status change                                                         | 10 checks                            |
| M11       | Org-wide and per-request audit history, cross-request isolation, resilience utility (4 unit-style checks)                    | 12 checks                            |
| M12       | Cross-org attack simulation (2 real orgs, forged bound arguments), rate limiting, security headers, full same-org regression | ~20 checks                           |
| M13       | Single continuous full-lifecycle run spanning every milestone's feature in one organization                                  | 32 checks                            |

**Total: 190+ individual scripted assertions across all milestones, all passing at the time of writing.**

**Test harness note:** several genuine test-script bugs were found and fixed along the way (ambiguous form-marker matching when a page has multiple similar forms; React's SSR comment-marker text interpolation breaking naive substring checks; forgetting to set a required form field). Each is called out in the relevant milestone's commit message where it could be mistaken for an app bug. Two real *application* bugs were also

found this way and are documented in Section 5 (the audit-log FK issue in M3, the scheduler status-sync issue in M6) alongside the Section 8 security finding.

## 11. Known Limitations & Deferred Scope

---

| Item                                                          | Why                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Excel (.xlsx) import not implemented — CSV only               | The only npm-available parser ( <code>xlsx</code> /SheetJS) carries unpatched high-severity vulnerabilities (prototype pollution, ReDoS) since SheetJS moved patched releases off the public registry. CSV covers the same functional need (FR-005) without that risk; revisit if a trustworthy parser becomes available.                           |
| Google Drive, WhatsApp, AI/OCR are mocked                     | No credentials available in this environment. See Section 9 for exactly what's needed to make each real.                                                                                                                                                                                                                                            |
| Rate limiting is process-local (in-memory)                    | Sufficient for a single-instance pilot (BR-13.2); would need a shared store (Redis, etc.) if ever deployed as multiple instances.                                                                                                                                                                                                                   |
| No automatic scheduled Collection Request creation            | Ch.5's "automatic" trigger for <i>starting</i> a new cycle each period wasn't built — Collection Requests are created manually (or would need a small addition to the scheduler once business-hours/frequency config per-service is defined). Everything <i>after</i> creation (conversation, documents, completion, reminders) is fully automatic. |
| "Documents in Processing" dashboard card typically reads zero | The mocked AI classification resolves synchronously, so there's never a persisted "processing" window to observe. Wired correctly; will show real data once/if an async OCR job queue exists.                                                                                                                                                       |
| Production deployment not executed                            | No hosting credentials provided; the app is deployment-ready (env var separation, health check, migration scripts, CRON_SECRET-gated endpoint) but was not pushed to a live host as part of this build.                                                                                                                                             |

## 12. Files Created / Changed

---

74 files changed since the initial landing-page commit (+19,617 / -4,053 lines). Organized by area:

### Database

`src/db/schema.ts` , `src/db/index.ts` , `drizzle/0000` – `0006` (7 migrations + snapshots),  
`drizzle.config.ts` , `scripts/migrate.ts` , `scripts/create-organization.ts`

### Auth & Security

`src/lib/auth/session.ts` , `password.ts` , `ratelimiter.ts` , `src/app/login/` (page, form, actions)

### Domain Logic

`src/lib/collectionRequestStateMachine.ts` , `conversationOrchestration.ts` , `scheduler.ts` ,  
`businessHours.ts` , `resilience.ts` , `audit.ts` , `csv.ts`

### AI (mocked)

`src/lib/ai/intentClassifier.ts` , `documentClassifier.ts`

### Storage (mocked)

`src/lib/storage/driveAdapter.ts`

### Data Access Layer

`src/lib/data/clients.ts` , `services.ts` , `collectionRequests.ts` , `dashboard.ts` , `auditLog.ts` ,  
`organizations.ts`

### App Routes & UI

`src/app/(app)/` — `layout.tsx` , `actions.ts` , and full subtrees for `dashboard/` , `onboarding/` ,  
`clients/` , `services/` , `collections/` , `audit/` , `settings/` ; `src/app/api/health/` ,  
`src/app/api/cron/tick/`

### Config

`next.config.ts` , `.env.example` , `package.json` (scripts)

### Untouched (verified via git diff)

`src/app/page.tsx` , `src/app/layout.tsx` , all of `src/components/landing/`

## 13. Recommended Next Steps

---

1. **Decide on real credentials** for Google Drive, WhatsApp Business API, and an AI/OCR provider — each swap is contained to one adapter file (Section 9), but Meta's Business verification in particular has its own lead time worth starting early.
2. **Choose a hosting target** and provision a real managed Postgres instance; the app already enforces this split (refuses to start in production without `DATABASE_URL` ) and ships a health-check endpoint for platform monitoring.
3. **Wire the external cron** (Vercel Cron, GitHub Actions scheduled workflow, or equivalent) to `POST /api/cron/tick` with the `CRON_SECRET` bearer token.
4. **Consider Excel import** once a trustworthy, actively-maintained parser is available, or if CSV-only proves insufficient for pilot firms in practice.
5. **Run the pilot** against Ch.20's success metrics (SM-20.1–20.4) with a real accounting firm once the above integrations are live — the dashboard, audit trail, and state machine are all in place to observe those metrics directly.

Centro Implementation Report — generated from the actual repository state (git history, file tree, and schema) at completion of milestone M13. All test results described were executed and observed, not projected.

## ADDENDUM

# Centro Implementation Report

## Epic 2 — Authentication & Onboarding Preparation

This addendum extends the Centro Implementation Report (Milestones M1–M13, dated July 19, 2026) with the work delivered in Epic 2. It is bound directly after the original report and should be read together with it. Two statements in the original report's Section 3.3 are superseded by this epic — see Section 14.2 below.

|                   |                                                                 |
|-------------------|-----------------------------------------------------------------|
| Prepared for      | noammaarachot@gmail.com                                         |
| Addendum covers   | EPIC 2 — Authentication & Onboarding Planning (Feature 1, 2, 3) |
| Addendum date     | July 19, 2026                                                   |
| Applies on top of | Centro Implementation Report, Milestones M1–M13 (bound below)   |
| Commit            | feat(auth): registration flow and onboarding preparation        |





## 14. Epic 2 — Authentication & Onboarding

### 14.1 Executive Summary

Epic 2 added self-service account registration, separated the landing page's lead-capture popup from the authenticated application, and put in place (without yet building the full wizard) the architecture for a first-time-onboarding gate. No visual/design-system change was made — every new form and page reuses the existing `Card` / `Button` / `TextField` primitives and copy tone introduced in the prior UI/UX redesign.

|                                               |                                                                 |                                                    |                                                                                       |
|-----------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------|
| NEW DB COLUMNS<br><b>4 (2 tables)</b>         | NEW ROUTES<br><b><code>/terms</code>, <code>/privacy</code></b> | RELOCATED ROUTE<br><b><code>/onboarding</code></b> | NEW SERVER ACTIONS<br><b><code>register</code>,<br/><code>finishOnboarding</code></b> |
| MIGRATION<br><b><code>drizzle/0007</code></b> |                                                                 |                                                    |                                                                                       |

### 14.2 Superseded Decisions from the Original Report

**Corrections to Section 3.3**

- "No public signup"** — the original report states organizations were provisioned only via the internal CLI script ( `npm run org:create` ), matching a sales-assisted, no-self-serve model. Epic 2 explicitly reverses this: `/login` now offers a "Create Account" tab that self-registers a brand-new organization. The CLI script still exists and still works (internal/pilot provisioning), but is no longer the only path.
- Onboarding location** — Section 6.1 lists `/onboarding` as a Sidebar-accessible page inside the authenticated app shell, always reachable. It has been relocated outside that shell (no longer in the Sidebar nav) and is now a first-login-only gated flow — see 14.6.

### 14.3 Database & Data Model Changes

Migration `drizzle/0007_gray_mentallo.sql`, applied via the existing `npm run db:migrate` tooling ( `src/db/index.ts`, unchanged). All four new columns are nullable — no backward-incompatible schema change.

| Table                      | Column                               | Purpose                                                                                                                                                                                                     |
|----------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>organizations</code> | <code>onboarding_completed_at</code> | Nullable timestamp. Same pattern as the existing <code>googleConnectedAt</code> / <code>whatsappConnectedAt</code> / <code>automationActivatedAt</code> columns. Drives the onboarding routing gate (14.6). |

|       |                     |                                                                                               |
|-------|---------------------|-----------------------------------------------------------------------------------------------|
| users | full_name           | Nullable — CLI-provisioned accounts still have none. Always set by self-service registration. |
| users | terms_accepted_at   | Nullable. Consent timestamp, set at registration.                                             |
| users | privacy_accepted_at | Nullable. Consent timestamp, set at registration.                                             |

**Backfill:** the same migration sets `onboarding_completed_at = COALESCE(automation_activated_at, now())` for every pre-existing organization, so pilot orgs created before Epic 2 are not sent back into the onboarding gate on their next login — only genuinely new organizations start ungated.

## 14.4 Registration Flow

`/login` now renders a tab switcher ( `src/app/login/AuthTabs.tsx` ) between the existing Login form and a new Create Account form ( `src/app/login/RegisterForm.tsx` ) inside the same card shell — no new route, so every existing `redirect("/login")` call across the codebase kept working unchanged.

- Fields: Full Name, Email, Password, Confirm Password — with show/hide toggles on both password fields.
- Two required checkboxes (Terms & Conditions, Privacy Policy), linking to new minimal placeholder pages ( `/terms` , `/privacy` — the landing Footer's legal links were previously `href="#"` ).
- Server-side validation ( `register()` in `src/app/login/actions.ts` ): required fields, email format and uniqueness, password minimum length (8), password/confirm match, both checkboxes accepted.
- On success: inserts a new `organizations` row and a new `users` row (password hashed with the existing script-based `hashPassword()` ), records two audit events ( `organization.created` , `employee.registered` , reusing `src/lib/audit.ts` ), creates a session ( `createSession()` , existing helper — the user is logged in immediately, never sent back to the login screen), then routes through the shared post-auth gate (14.6).

### The organization name is a temporary placeholder, not a product decision

Registration deliberately does not ask for an office/company name — only Full Name, Email, Password, Confirm Password, per spec. Since `organizations.name` is `NOT NULL` , it is filled with the registrant's full name purely to satisfy that constraint. The real office name is planned as onboarding-wizard step 1 in the next epic; nothing in this epic treats the placeholder as final, and no UI surface introduced in Epic 2 displays it as "your organization name."

## 14.5 Login Flow (Updated)

`login()` is otherwise unchanged (same rate limiting, same credential check, same audit event) — only its final step changed: instead of a hardcoded `redirect("/dashboard")` , it now calls the same shared gate function registration uses (14.6), so both entry points make the exact same onboarding-vs-dashboard decision through one function, not two separately-maintained redirects.

## 14.6 First-Time User / Onboarding Architecture

New module `src/lib/onboarding.ts` is the single source of truth for the onboarding gate:

| Export                                              | Role                                                                                                                                                                                                                                  |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>needsOnboarding(organization)</code>          | The one predicate: <code>!organization.onboardingCompletedAt</code> .                                                                                                                                                                 |
| <code>redirectAfterAuth(organizationId)</code>      | Loads the organization, applies <code>needsOnboarding()</code> , and redirects to <code>/onboarding</code> or <code>/dashboard</code> . Called as the last step of both <code>login()</code> and <code>register()</code> .            |
| <code>markOnboardingComplete(organizationId)</code> | Sets <code>onboardingCompletedAt = now()</code> . Called from the "continue to dashboard" action and from <code>activateAutomation()</code> — the seam the real onboarding wizard (next epic) will also call from its own final step. |

`src/app/(app)/layout.tsx` — the shared layout every Dashboard/Clients/Services/Collections/Settings/Audit route renders through — also calls `needsOnboarding()` as a defensive backstop (e.g. a bookmarked `/dashboard` URL from before onboarding finished), reusing the same predicate rather than re-implementing the condition anywhere else.

**What "Office Setup" is today, and what it becomes:** the page previously reachable from the Sidebar as "הקמת המערכת" (Google/WhatsApp connect toggles, CSV client import, automation activation — all built in M4 of the original report) was moved, content and logic untouched, from `src/app/(app)/onboarding/` to `src/app/onboarding/` (same URL — Next.js route groups don't affect the path, so no `redirect()` string anywhere needed to change). It's no longer a Sidebar nav item; it is now what a first-time user lands on immediately after registering or logging in, and it's the placeholder the real onboarding wizard replaces in the next epic. One new action, `finishOnboarding()` , adds an explicit "Continue to Dashboard" exit so an organization is never stuck there just because it isn't ready to activate automation yet.

## 14.7 Landing Page / Application Separation

**Root cause:** `DemoRequestModal` (the 45-second lead-capture popup) was mounted in `src/app/layout.tsx` — the root layout every route in the app renders through, authenticated or not — so its internal timer could fire on `/login` , `/onboarding` , or any Dashboard-family route.

**Fix:** moved the `<DemoRequestModal />` mount from the root layout into `src/app/page.tsx` (the actual public landing page) alongside its other sections. It is no longer part of the module graph for any other route, so it cannot mount there. `AccessibilityProvider` / `AccessibilityWidget` / `MotionProvider` are unrelated to this popup and were left in the root layout unchanged.

## 14.8 Routes & Server Actions — Delta on Section 6

| Route               | Change                                                                                                                                                                  |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>/login</code> | Now renders both Login and Create Account (tabbed); redirects an already-authenticated visitor through the shared gate instead of straight to <code>/dashboard</code> . |

|                                                |                                                                                                                      |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <code>/onboarding</code>                       | Moved out of the <code>(app)</code> route group (no Sidebar); gated to first-time users instead of always reachable. |
| <code>/terms</code> ,<br><code>/privacy</code> | New. Minimal placeholder pages linked from the registration checkboxes.                                              |

| File                                       | Actions added                                                                                                                   |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <code>src/app/login/actions.ts</code>      | <code>register</code> (new); <code>login</code> now ends with <code>redirectAfterAuth()</code> instead of a hardcoded redirect. |
| <code>src/app/onboarding/actions.ts</code> | <code>finishOnboarding</code> (new); <code>activateAutomation</code> now also calls <code>markOnboardingComplete()</code> .     |

## 14.9 Testing Performed

Same real-HTTP, scripted-E2E standard as M1–M13: a Playwright script drove an actual browser against the running dev server (temporarily installed for the test run, removed afterward — not a project dependency). All of the following were observed passing, not just asserted:

- Register a new account → lands on `/onboarding` directly (not `/dashboard` first) → Sidebar absent on that page → "Continue to Dashboard" → lands on `/dashboard` .
- Log out, log back in with the same (now-onboarded) account → lands straight on `/dashboard` , confirming the gate persists correctly across sessions.
- Log in with a pre-Epic-2 account (created before this migration) → lands straight on `/dashboard` , confirming the backfill in 14.3.
- Duplicate-email registration and password-mismatch submissions both surface the correct field-level error and do not create a row.
- 46-second real-time wait on an authenticated `/dashboard` session confirmed the demo popup never appears.
- `npm run build` and `npm run lint` both clean.

## 14.10 Known Limitations & Next Steps (Epic 2)

| Item                              | Why / Next step                                                                                                                                                                                                                                         |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Onboarding wizard not built       | Explicitly out of scope for this epic — Feature 3 asked only for the routing/gating architecture to be ready. The relocated Office Setup page is the interim placeholder; <code>markOnboardingComplete()</code> is the seam the real wizard calls next. |
| Organization has no rename UI yet | The placeholder name set at registration (14.4) can't currently be edited anywhere in the app (Settings only covers business hours). Real name collection is planned as the wizard's first step.                                                        |

|                                                                |                                                                                                                                                                                       |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>/terms</code> and <code>/privacy</code> are placeholders | No legal copy was authored as part of this epic — the pages exist so the required checkboxes link to something real, not to publish final legal text.                                 |
| CLI provisioning script unchanged                              | <code>scripts/create-organization.ts</code> still works as before for internal/pilot use and does not collect a full name — those accounts keep <code>users.full_name = NULL</code> . |

---

Epic 2 addendum — generated from the actual repository state (git history, schema, and a live scripted test pass) at completion of the Authentication & Onboarding Planning epic. All test results described were executed and observed, not projected.

## ADDENDUM

# Centro Implementation Report

## Epic 3 — Premium First-Time Onboarding Wizard

This addendum extends the Centro Implementation Report (Milestones M1–M13) and the Epic 2 addendum (Authentication & Onboarding Preparation) with the work delivered in Epic 3. It is bound directly after both and should be read together with them.

|                   |                                                                                                       |
|-------------------|-------------------------------------------------------------------------------------------------------|
| Prepared for      | noammaarachot@gmail.com                                                                               |
| Addendum covers   | EPIC 3 — Premium First-Time Onboarding Wizard (9 steps)                                               |
| Addendum date     | July 19, 2026                                                                                         |
| Applies on top of | M1–M13 report + Epic 2 addendum (bound before this section)                                           |
| Commit            | <code>feat(onboarding): premium multi-step onboarding wizard with business-type classification</code> |



## 15. Epic 3 — Premium First-Time Onboarding Wizard

### 15.1 Executive Summary

Epic 3 replaces the interim "Office Setup" page (built in Epic 2 as an explicit placeholder) with a real 9-step onboarding wizard at `/onboarding`: Welcome, Office Information, Connect Services, Import Clients, AI Client Analysis, Configure Required Documents, Reminder Rules, Summary, and Completion. It introduces a new product concept, **Business Type**, and a mocked AI classifier that sorts imported clients into it automatically — while deliberately reusing the existing Service / Collection-Request engine underneath rather than building a second, parallel system.

|                                     |                                             |                                        |                                  |
|-------------------------------------|---------------------------------------------|----------------------------------------|----------------------------------|
| WIZARD STEPS<br><b>9</b>            | NEW DB OBJECTS<br><b>1 table, 7 columns</b> | NEW STARTER BUSINESS TYPES<br><b>5</b> | MIGRATION<br><b>drizzle/0008</b> |
| REAL BUGS FOUND & FIXED<br><b>2</b> |                                             |                                        |                                  |

### 15.2 Architectural Decision: Business Type = a Product Concept, Not a New Domain Model

The central design question was how "Business Type" (Step 5's classification, Step 6's document lists, Step 7's reminder rules) relates to the pre-existing `services` / `service_document_requirements` engine, which already models "a named thing that owns a document-requirement list." The chosen approach, confirmed with the user before implementation: **Business Type is a UI/product concept only** — every business type is backed by exactly one auto-managed Service of the same name. Users never see the word "Service" anywhere in the wizard; the existing `/services` pages (untouched by this epic) continue to work as the already-built editor for those same rows. This reused 100% of the Collection Request engine, the requirement-snapshot logic, and the dashboard/audit trail, with zero new parallel machinery.

### 15.3 Database & Data Model Changes

Migration `drizzle/0008_cold_bucky.sql`. All additions nullable/backward compatible.

| Table                                             | Column(s)                                                                                                                                                               | Purpose                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>business_types</code><br><i>(new table)</i> | <code>id</code> , <code>organizationId</code> , <code>name</code> , <code>serviceId</code> ,<br><code>isCustom</code> , <code>createdAt</code> , <code>updatedAt</code> | Org-scoped catalog (never a shared global list, so each firm can freely rename/add/ remove its own types). <code>serviceId</code> is the 1:1 link to the backing Service. Seeded with five starter types (Limited Company, Sole Proprietor, Exempt Business, Nonprofit, Payroll Only) the first time an org reaches Step 5 — the |



|               |                                                                                                                                                                      |                                                                                                                                                                                                                                                                                               |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               |                                                                                                                                                                      | starter list itself is a code-level default, like any product's recommended template, but every row it produces is a real, independently-editable DB row from that point on.                                                                                                                  |
| clients       | business_type_id                                                                                                                                                     | Nullable FK. Null = "Unclassified." Set by the mocked AI classifier or manually via Step 5's bulk-assign flow.                                                                                                                                                                                |
| services      | reminder_interval_days_override,<br>inactivity_timeout_minutes_override,<br>business_hours_start_override,<br>business_hours_end_override,<br>business_days_override | Five nullable per-Business-Type overrides mirroring organizations ' equivalent columns exactly. Null means "use the organization's default." Every pre-Epic-3 service has all five null.                                                                                                      |
| organizations | logo_url, onboarding_step                                                                                                                                            | logo_url : a small logo stored directly as a data: URL (no blob storage exists in this project — a real, working solution at this size, not a mock).<br>onboarding_step : which step (1–9) a first-time user last reached, so closing the tab mid-wizard resumes there instead of restarting. |

## 15.4 Reminder Rules Are Real Per-Business-Type Overrides

Confirmed with the user as a deliberate, real (not cosmetic) change: reminder interval, inactivity timeout, and business hours/days can now differ per Business Type, not just per organization. `src/lib/businessHours.ts` gains `resolveScheduleConfig(organization, service)` — the single place that resolves the effective five-field config (service override ?? organization default). `src/lib/conversationOrchestration.ts`'s `sendOutboundMessage` (the one gating point for every automated send) now resolves the sending conversation's Collection Request → Service before checking business hours, instead of reading the organization alone. `src/lib/scheduler.ts`'s two queries (idle-open, stale-waiting) now join through to `services` and resolve each conversation's effective cutoff individually in application code, rather than computing one shared SQL cutoff per organization. Both changes are additive: a service with no overrides resolves to exactly the pre-Epic-3 organization-wide values, so the M1–M13 scheduler/ conversation test coverage's expected behavior is unchanged for all pre-existing data.

## 15.5 Mocked AI: Client Classification

`src/lib/ai/businessTypeClassifier.ts` follows the same "real interface, mocked transport" pattern already established by `src/lib/ai/documentClassifier.ts` (M9): no LLM credential is configured for this pilot, so classification is deterministic — an explicit business-type column in the imported file (real, office-provided data) takes priority when present; otherwise a small set of Hebrew/English legal-suffix heuristics apply

(`"/Ltd"` → Limited Company, `עמותה` → Nonprofit, `עוסק פטור` → Exempt Business, etc.); anything unmatched is left Unclassified rather than guessed. Swapping in a real LLM later touches only this file.

## 15.6 Import: "Excel / CSV" Behind an Adapter

M4 deliberately shipped CSV-only because the only npm .xlsx parser (xlsx/SheetJS) carries unpatched high-severity vulnerabilities on the public registry. Per explicit direction for this epic, the wizard's Step 4 still always labels itself "Import Excel / CSV" and accepts `.csv/.xlsx/.xls` by file picker, but the actual parsing is routed through a new `src/lib/import/clientImportAdapter.ts`: `.csv` is fully real (reuses the existing hand-written parser unchanged); `.xlsx / .xls` are recognized and rejected with a friendly "export as CSV" message instead of a crash. Native Excel parsing — if a trustworthy package becomes available — is a change contained to this one adapter file; nothing in the wizard, its actions, or its UI needs to change.

## 15.7 The Wizard's Architecture

`/onboarding` is **searchParam-driven** (`?step=1 ... 9`), the same pattern already used elsewhere in this app (e.g. the Dashboard's `?queue=` drill-down) — not a client-side single-page wizard with its own state machine. Each step is a focused Server Component in `src/app/onboarding/steps/`, rendered by a `page.tsx` switch statement that loads exactly the data that step needs; a shared `WizardShell.tsx` provides the progress bar, title/description, a reusable `HelpTip` "Why is this important?" popover (`src/components/app/HelpTip.tsx`, available to any future feature, not onboarding-only), and the Previous link. Inserting a new step in a future epic is a one-file-plus-one-switch-case change, not a rewrite — the explicit "modular, extensible" requirement for this epic.

Every step persists through the existing server-action convention (plain `<form action={...}>`, no client-side wizard state to keep in sync) — `src/app/onboarding/actions.ts` keeps Epic 2's

`connectGoogle` / `connectWhatsapp` / `finishOnboarding` etc. entirely unchanged in behavior and reuses them directly (Step 3, Step 9); Step 6 reuses the existing `addRequirement` / `removeRequirement` actions from `services/actions.ts` unmodified in logic (only extended with an optional `returnTo` field so the wizard can reuse them without navigating away to `/services/[id]`).

### Real bug found & fixed #1 — CSV parser and Hebrew legal suffixes

Testing with a realistic Hebrew client-name fixture (e.g. a company named with the extremely common `בע"מ` suffix) surfaced a real, pre-existing parsing bug in `src/lib/csv.ts` (written in M4): the hand-written parser toggled "inside a quoted field" on *any* `"` character, including one appearing mid-field in an otherwise-unquoted value — exactly what `בע"מ` contains. This silently corrupted the rest of the file into one swallowed field, losing most or all rows. Fixed to match RFC 4180 correctly: a `"` only starts a quoted field when it is the field's very first character. This is a real fix to existing import logic, not new code path — every future CSV import (wizard or otherwise) benefits.

### Real bug found & fixed #2 — same-page Server Action redirects

Several wizard steps (and, once discovered, two pre-existing forms — Settings' business-hours form and `/services/[id]`'s requirement add/remove) submit a Server Action that redirects back to the exact same page it was submitted from. Next.js's client router does not reliably treat a `redirect()` to an identical URL as "new data available," so the mutation was persisted correctly to the database but the visible page kept showing pre-mutation data until a manual reload. Fixed using Next.js 16's `refresh()` API (`next/cache` — see `node_modules/next/dist/docs/.../functions/refresh.md`), which exists specifically for this case: same-page mutations now call `refresh()` and simply return, with no `redirect()` at all (the two aren't meant to compose in the same call); genuine cross-page navigations were left as plain `redirect()` calls, unaffected. Verified with a real scripted browser test asserting the updated content is visible without a manual reload, not just that the database row changed.

## 15.8 Step-by-Step Reference

| Step                            | What it does                                                                                                                                                                                   | Persists via                                                                |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| 1. Welcome                      | Greeting, estimated time, no data entry.                                                                                                                                                       | <code>advanceOnboardingStep(2)</code>                                       |
| 2. Office Information           | Office name (replaces the Epic-2 registration placeholder) + optional logo (client-side canvas resize → data URL).                                                                             | <code>updateOfficeInfo</code> — shared with Settings, see 15.9              |
| 3. Connect Services             | Google Drive / WhatsApp Business toggles with plain-language "why" copy. Not a hard gate — BR-001's real activation gate still lives at completion (15.9).                                     | Epic 2's existing <code>connectGoogle</code> / <code>connectWhatsapp</code> |
| 4. Import Clients               | Excel/CSV upload → adapter → insert (duplicate-phone-aware, same as before) → classify.                                                                                                        | <code>importAndClassifyClients</code>                                       |
| 5. AI Client Analysis           | Clickable per-type count cards with drill-down; "Assign Business Type" bulk flow for Unclassified clients (multi-select + existing-or-new-type picker) — no spreadsheet editing ever required. | <code>assignBusinessTypeAction</code>                                       |
| 6. Configure Required Documents | Per business type: existing requirements (removable), suggested-but-not-added quick-adds, free-text custom document add.                                                                       | reused <code>addRequirement</code> / <code>removeRequirement</code>         |
| 7. Reminder Rules               | Per business type: a "customize for this type" toggle + working days/hours/reminder interval/inactivity timeout, prefilled with the effective (override ?? org default) values.                | <code>updateServiceScheduleOverrides</code>                                 |
| 8. Summary                      | Read-only recap, always derived live from real counts (no separate "summary state" to go stale).                                                                                               | —                                                                           |

|               |                                                                                                                                                         |                                                                      |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| 9. Completion | Best-effort activates automation if both integrations were connected (BR-001 gate, unchanged logic), marks onboarding complete, redirects to Dashboard. | Epic 2's existing <code>finishOnboarding</code> (extended, see 15.9) |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|

## 15.9 Post-Wizard Editability

Every setting the wizard configures remains editable afterward, per explicit requirement:

- **Office name/logo** — new `OfficeInfoForm` component ( `src/components/app/OfficeInfoForm.tsx` ) shared by Step 2 and a new section on `/settings` ; both submit through the same `updateOfficeInfo` action.
- **Document requirements per business type** — already fully editable via the pre-existing `/services/[id]` page (a business type *is* that service); no new UI needed.
- **Reminder overrides per business type** — new `ServiceScheduleOverrideCard` component ( `src/components/app/ServiceScheduleOverrideCard.tsx` ) shared by Step 7 and a new card on `/services/[id]` .
- **Automation on/off** — `activateAutomation` / `deactivateAutomation` (Epic 2, unchanged core logic) now surface as a toggle on `/settings` instead of only inside the old single-page Office Setup — the wizard itself no longer shows this control, only Step 9's best-effort auto-activation.

## 15.10 Testing Performed

Same real-HTTP, scripted-browser standard as prior epics (Playwright, temporarily installed for the run, removed afterward). A single continuous pass: register a fresh organization → walk all 9 steps (real CSV fixture with a deliberate mix of classifiable and unclassifiable Hebrew business names) → verify AI analysis counts, bulk-reclassification, custom document addition, and reminder-override saving are all *visibly* reflected without a manual reload → finish → confirm `/services` lists the five business-type-backed services with their configured requirements → confirm the new Reminder Overrides and Settings sections render → log out and back in as the same (now-onboarded) org → confirm it lands straight on `/dashboard` , not back in the wizard → log in as a pre-Epic-3 account → confirm unaffected. `npm run build` and `npm run lint` both clean.

## 15.11 Known Limitations & Next Steps (Epic 3)

| Item                                                  | Why / Next step                                                                                                                                                  |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Classification heuristics are a small, fixed rule set | Sufficient for a mocked pilot; a real LLM classifier (already interface-compatible) would handle far more name patterns and languages. See 15.5.                 |
| Native .xlsx/.xls parsing not implemented             | Same documented security tradeoff as M4 — the adapter (15.6) is ready for a trustworthy parser whenever one becomes available.                                   |
| Logo storage is a data: URL, not a file host          | Fine at the size this is capped to (client-side resized before upload); a real asset host would be needed for larger images or multiple assets per organization. |

|                                                              |                                                                                                                                                               |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Custom Business Types have no suggested requirement template | Only the five starter types ship with a pre-defined checklist (15.3) — an office-created custom type starts empty, filled entirely via "Add Custom Document." |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|

---

Epic 3 addendum — generated from the actual repository state (git history, schema, and a live scripted end-to-end test pass) at completion of the Premium First-Time Onboarding Wizard epic. All test results and both bug findings described were executed and observed, not projected.

## ADDENDUM

# Centro Implementation Report

## Production-Ready Client Import: Native Excel, Hebrew Business-Type Recognition & Localization

This addendum extends the Centro Implementation Report and its Epic 2/3 addenda with a follow-up hardening pass on the onboarding wizard's client-import step, prompted by real testing against an actual Israeli accounting office's file.

|                   |                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------------|
| Prepared for      | noammaarachot@gmail.com                                                                                             |
| Addendum covers   | Native .xlsx support, deterministic Hebrew business-type mapping, full Hebrew localization of the onboarding wizard |
| Applies on top of | M1–M13 report + Epic 2 & Epic 3 addenda (bound before this section)                                                 |



# 16. Client Import Hardening — Native Excel, Business-Type Recognition, Localization

## 16.1 Executive Summary

Epic 3 shipped a working CSV-only import with a rejected-with-a-friendly-message stub for .xlsx, and English canonical names for the five starter Business Types. Real testing surfaced three concrete gaps: offices work in Excel, not CSV, by default; the classifier's synonym list didn't cover the actual Hebrew terminology Israeli accounting files use; and the wizard showed those English internal names directly to Hebrew-speaking users. All three are fixed below, plus a fourth, related fix: the classifier function is now clearly structured as a deterministic match first, with the (currently unconfigured) AI path explicitly gated to run only after deterministic matching has failed — matching the intended architecture, not just its label.

|                                              |                                                     |                                         |                                       |
|----------------------------------------------|-----------------------------------------------------|-----------------------------------------|---------------------------------------|
| NEW DEPENDENCY<br><b>exceljs (read-only)</b> | NEW VULNERABILITIES<br>ADDED<br><b>0 (verified)</b> | BUSINESS-TYPE<br>SYNONYMS<br><b>30+</b> | AUTOMATED TESTS<br>ADDED<br><b>28</b> |
| TEST RUNNER<br><b>vitest (new)</b>           |                                                     |                                         |                                       |

## 16.2 Native Excel (.xlsx) Support

**Library selection process** (documented per the explicit requirement to research rather than assume): the SheetJS `xlsx` package — the option historically rejected in M4/Epic 3 — remains at npm version 0.18.5, years stale, because SheetJS stopped publishing patched releases to the public registry. `exceljs` was evaluated instead: installed and checked with `npm audit` before being accepted, not after. The install itself added zero new advisories; a subsequent full `npm install` surfaced one transitive finding — `exceljs` pins `uuid@8.3.2`, and that major version has a published moderate advisory (GHSA-w5hq-g745-h8pq, a missing bounds check reachable only when a caller supplies a custom buffer to `uuid`'s `v3/v5/v6` functions — not a pattern this codebase's read-only usage triggers, but not accepted regardless). Fixed with an `npm overrides` entry in `package.json` pinning `uuid` to `^11.1.1` across the whole tree; verified via `npm audit` and a read/write round-trip smoke test afterward. The project's final audit count (6 moderate findings, all pre-existing in `drizzle-kit`'s esbuild toolchain and Next's `postcss`, unrelated to this change) did not increase.

**Usage is deliberately narrow:** only `Workbook.xlsx.load()` and worksheet/row iteration are called (`src/lib/import/xlsxParser.ts`) — `exceljs`'s write and streaming APIs are never invoked anywhere in this codebase, keeping the exposed surface to exactly "parse an untrusted spreadsheet into strings," regardless of what else the package can do.

**Pipeline reuse:** `src/lib/csv.ts` was split into `parseCsv()` (format-specific: turns CSV text into a raw `string[][]` grid) and the new `mapRowsToClientRows()` (format-agnostic: turns that grid into `{ name, phone,`



email, notes, businessType } via the Hebrew/English header-alias table). `xlsxParser.ts` produces the same `string[][]` shape from a worksheet, so it calls the exact same `mapRowsToClientRows()` — column detection, required-field validation, and business-type-column recognition are implemented exactly once for both formats, not duplicated. `src/lib/import/clientImportAdapter.ts` dispatches by file extension; legacy binary `.xls` (a structurally unrelated, pre-2007 format with no comparably safe actively-maintained parser available) is explicitly out of scope and rejected with a message naming the two formats that are supported, rather than a raw parse error.

### 16.3 Deterministic Hebrew Business-Type Recognition

**Root cause of "detected all clients but classified none"**

The classifier logic itself was already capable of matching terms like **עוסק פטור** — the actual gap was coverage and structure: the synonym list was small (only the full legal phrases, not the shorthand terms real files use, e.g. **מורשה** alone rather than always **עוסק מורשה**), and it lived as ad-hoc regex inside the classifier rather than as an explicit, reviewable dictionary. Rebuilt as `BUSINESS_TYPE_SYNONYMS` in `src/lib/ai/businessTypeClassifier.ts` — an explicit canonical-name-to-synonym-list map, matched by normalized substring containment (longest synonym first, so a specific phrase is preferred over a word that's also a substring of it), covering every example given plus the common Hebrew/English variants around it.

| Canonical (Hebrew, as stored) | Recognized synonyms (subset)                             |
|-------------------------------|----------------------------------------------------------|
| חברה בע"מ                     | plus Ltd/Inc/Corp/Company — בע"מ, בעמ, חברה              |
| עוסק מורשה                    | plus sole proprietor/self-employed — עוסק/ת מורשה, מורשה |
| עוסק פטור                     | plus exempt/exempt business — עוסק/ת פטור, פטור          |
| עמותה                         | plus nonprofit/NGO/amuta — עמותת                         |
| שכר בלבד                      | plus payroll/payroll only — שכר                          |

**Priority order, made explicit in code and its doc comment** ( `classifyClientBusinessType` ):

- Deterministic dictionary match against an explicit business-type column, when the import file provides one (real, office-supplied data) — confidence 1.0.
- AI fallback** — a dedicated `classifyViaAI()` function, called only if step 1 found nothing. No LLM/OCR provider is configured for this pilot (same mock-first decision as `documentClassifier.ts` ), so this is currently a documented no-op returning `null` — not a fake guess — and wiring in a real provider later touches only this one function, which by construction cannot run before deterministic matching has had its chance.
- Deterministic dictionary match against the client's own *name* (some offices embed the type there instead of a separate column) — lower confidence, last resort.

4. Unclassified — always available for manual assignment in Step 5, never silently guessed.

The business-type column's own header detection was also broadened ( `src/lib/csv.ts` 's `HEADER_ALIASES` ) to recognize more realistic real-world column names: **מעמד, מעמד, ישות משפטית, סוג ישות, סיווג, סוג עסק, סטטוס עסק**, plus the English equivalents.

## 16.4 Full Hebrew Localization

The five starter Business Types' canonical *stored* names ( `src/lib/businessTypes.ts` 's `STARTER_BUSINESS_TYPES` ) were changed from English to Hebrew directly — **חברה בע"מ, עוסק מורשה, עוסק, פטור, עמותה, שכר בלבד** — rather than adding a separate display-translation layer. Because every wizard step (5, 6, 7, 8) and the post-wizard `/services` page already render this field dynamically rather than hardcoding labels, this single seed-data change is what makes the whole experience Hebrew, with no risk of a translation layer drifting out of sync with the underlying data. The suggested document-requirement names seeded per type (e.g. "Expense Invoices") were translated to their standard Hebrew accounting terms (**חשבוניות הוצאה**, etc.) for the same reason — those are also rendered directly in Step 6. Everything internal that isn't user-facing (event-type strings, code identifiers, comments) intentionally stayed in English, consistent with the rest of this codebase.

**Note on existing data:** organizations that completed onboarding before this change have business types already seeded with the old English names — this only affects the local pilot/dev database (no production tenants exist yet), and every name remains fully editable per-organization from Step 6 or `/services` regardless.

## 16.5 Post-Import UX

Step 5's top banner was restructured into an explicit three-number success summary — clients imported, classified automatically, and requiring manual classification — replacing the single "N clients found" line. This sits directly above the existing per-type drill-down cards and the "Assign Business Type" bulk-classification flow for anything left unclassified, so the full picture (what happened, and what — if anything — still needs a decision) is visible in one place immediately after upload.

## 16.6 Automated Testing

No test runner existed in this project before this change. Added `vitest` (zero new advisories, minimal config — no DOM/plugin dependency needed since these are unit tests over parsing/matching logic, not component rendering) with `npm run test`. 28 tests across four files: `csv.test.ts` (RFC 4180 edge cases including the Epic-3 **בע"מ** mid-field-quote fix, Hebrew header aliasing, the required-column error path), `xlsxParser.test.ts` (a real exceljs-generated workbook parsed back, numeric-cell stringification, blank-row skipping, Hebrew round-tripping), `clientImportAdapter.test.ts` (end-to-end dispatch for both real formats plus the `.xls/unknown-extension` rejection paths), and `businessTypeClassifier.test.ts` (every synonym example from this requirement, the name-fallback path, and the "don't match a type the org never created" guard).

## 16.7 Files Touched

| File                                       | Change                                                                                          |
|--------------------------------------------|-------------------------------------------------------------------------------------------------|
| package.json                               | + exceljs , + vitest (dev), overrides.exceljs.uuid , test script                                |
| src/lib/csv.ts                             | split into parseCsv() + mapRowsToClientRows() ; broadened business-type header aliases          |
| src/lib/import/xlsxParser.ts               | <b>new</b> — native .xlsx reading                                                               |
| src/lib/import/clientImportAdapter.ts      | .xlsx now real; .xls/unknown messaging updated                                                  |
| src/lib/ai/businessTypeClassifier.ts       | rebuilt around an explicit synonym dictionary + priority-ordered, AI-gated-as-fallback matching |
| src/lib/businessTypes.ts                   | starter type + requirement names translated to Hebrew                                           |
| src/app/onboarding/actions.ts              | await on the now-async classifier; richer audit-event metadata                                  |
| src/app/onboarding/steps/Step5Analysis.tsx | explicit imported/classified/needs-review success summary                                       |
| 4 new *.test.ts files, vitest.config.ts    | automated test suite                                                                            |

Addendum generated from the actual repository state (git history, a full npm audit pass, and the automated + scripted browser test results described above) after this hardening pass. All findings and test results were executed and observed, not projected.

## ADDENDUM

# Centro Implementation Report

## Business-Type Data Migration: Upgrading Pre-Existing Organizations to Hebrew

This addendum documents a bug found during real-user re-testing of the previous (Import Hardening) addendum: organizations seeded before the English→Hebrew business-type rename were never upgraded, so both the wizard's display and client classification silently kept failing for them. Fixed with an automatic data migration, not a fresh-database requirement.

|                   |                                                                                       |
|-------------------|---------------------------------------------------------------------------------------|
| Prepared for      | noammaarachot@gmail.com                                                               |
| Addendum covers   | Root cause analysis, the data migration, and end-to-end re-verification               |
| Applies on top of | M1–M13 report + Epic 2, Epic 3 & Import Hardening addenda (bound before this section) |



## 17. Business-Type Data Migration — Upgrading Existing Organizations

### 17.1 Bug Report

The previous addendum's own "Note on existing data" flagged, but did not act on, a gap: it stated the English→Hebrew rename "only affects the local pilot/dev database" and every name "remains fully editable." Real re-testing proved that caveat insufficient — the reporter's own test organization still showed the old English labels (Limited Company, Sole Proprietor, Exempt Business, Payroll Only, Nonprofit) throughout the onboarding UI, and a 20-client import file with a populated Business Type column produced **zero** automatic classifications — every client fell through to "requires manual classification," despite the classifier logic itself being correct and already covered by 28 passing unit tests.

### 17.2 Root Cause

#### One root cause, two visible symptoms

`seedStarterBusinessTypes()` is deliberately idempotent-by-name: it only creates starter types an organization doesn't already have, so that a firm which has already customized its checklist is never silently overwritten. That guarantee has a corollary its author didn't account for: when the starter list's canonical names changed from English to Hebrew, organizations that had already been seeded under the old names were *never revisited* — the idempotency check saw "this org already has a type named 'Limited Company'" and correctly left it alone, because at the code level nothing distinguishes "the user renamed this on purpose" from "this is stale seed data."

That single stale row set explains both symptoms:

1. **English UI** — Steps 5–7 and `/services` render `business_types.name` / `services.name` directly; a stale org's rows were still English.
2. **Zero classification** — `classifyClientBusinessType()` resolves an explicit "business type" column value (e.g. `חברה בע"מ`) to its Hebrew canonical name via `BUSINESS_TYPE_SYNONYMS`, then calls `findCandidateByName()` against the organization's actual `business_types` rows. For a stale org those rows were still named "Limited Company" etc. — an exact-string lookup for the Hebrew name found no candidate, so every classification attempt silently fell through to Unclassified, regardless of a correctly populated source column.

Confirmed empirically before writing any fix, not assumed: a direct query of every local organization's `business_types` rows showed 2 of 6 still on the old English names — including an organization named `רואי חשבון` (a genuine Hebrew accounting-firm name), matching the reporter's own account exactly.

### 17.3 Fix: An Automatic Upgrade Migration

Per the explicit requirement — *upgrade existing installations automatically instead of requiring a fresh database* — the fix is a proper, hand-authored Drizzle data migration (`drizzle/0009_hebrew_business_type_names.sql`), generated via `drizzle-kit generate --custom` so it is correctly registered in `drizzle/meta/_journal.json` and runs through the exact same `npm run db:migrate` pipeline as every other migration — not a one-off script that would only help this one machine.

The migration, scoped conservatively in three layers:

1. Renames `business_types.name` from each of the five original English defaults to its Hebrew canonical equivalent, matched by exact old name — an organization that already renamed a type to something else entirely is left untouched.
2. Renames the matching `services.name` row (the backing service is what `/services` and Collection Requests actually display), same exact-match scoping.
3. Renames the standard document-requirement labels those five services were originally seeded with (e.g. "Expense Invoices" → **חשבוניות הוצאה**), scoped by `service_id` IN (`SELECT id FROM services WHERE name` IN (`<the Hebrew names just written above>`)) — so a requirement a user manually added to some unrelated custom service under a coincidentally matching English label is never touched.

**PGLite constraint discovered while authoring this migration**

PGLite (and the underlying Postgres protocol) cannot prepare more than one SQL command per statement. Drizzle's migrator splits a migration file into separate prepared statements on `--> statement-breakpoint` markers, so every individual `UPDATE` needs its own marker — batching several statements under one shared marker throws `error: cannot insert multiple commands into a prepared statement` (Postgres 42601). The final migration has 15 separate breakpoints, one per statement.

17.4 Verification

**Confirmed via direct DB inspection, then a full scripted re-run of the wizard**

After applying the migration, all 6 local organizations — including the previously-stale **רז רואי חשבון** — were re-queried directly: every `business_types`, `services`, and `service_document_requirements` row is now correctly Hebrew.

A full, fresh browser-driven pass of the onboarding wizard was then run end-to-end (registration → Step 1 through Step 5) against a new organization, importing a 20-row CSV fixture deliberately built to exercise every synonym-matching path: 5 rows using the exact Hebrew canonical terms, 5 using English/abbreviated synonyms (Ltd, Sole Proprietor, Nonprofit, ...), 5 using short Hebrew shorthand (**amuta**, **payroll**, **פטור**, **מורשה**, **בע"מ**), and 5 with an *empty* business-type column to exercise the name-based fallback and the final Unclassified path.

|       |        |
|-------|--------|
| Check | Result |
|-------|--------|

|                                                         |                                                                                                                      |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| English business-type strings anywhere in the Step 5 UI | 0 (none found)                                                                                                       |
| Expected Hebrew canonical names rendered                | All 5 present (חברה בע"מ, עוסק מורשה, עוסק פטור, עמותה, שכר בלבד)                                                    |
| Clients imported                                        | 20                                                                                                                   |
| Auto-classified                                         | 19 (4 / 4 / 4 / 4 / 3 across the five types)                                                                         |
| Requiring manual classification                         | 1 — a name with no recognizable business-type signal at all (correctly deferred to manual assignment, never guessed) |

This matches the deterministic classifier's documented behavior exactly: every row with a recognizable signal (explicit column or embedded in the name) classified correctly at confidence 1.0 or 0.6 respectively; the one row with neither correctly fell through to Unclassified rather than being silently mis-guessed.

Full regression pass after the fix: `npm run build` (all 17 routes compiled), `npm run lint` (clean), and `npm run test` (28/28 passing, unchanged) — the migration is data-only and touches no schema or application code, so no existing test coverage was expected to move, and none did.

### 17.5 Files Touched

| File                                                     | Change                                                                              |
|----------------------------------------------------------|-------------------------------------------------------------------------------------|
| <code>drizzle/0009_hebrew_business_type_names.sql</code> | <b>new</b> — hand-authored data-only migration upgrading pre-existing organizations |
| <code>drizzle/meta/_journal.json</code>                  | registers migration 0009                                                            |
| <code>drizzle/meta/0009_snapshot.json</code>             | <b>new</b> — unchanged schema snapshot (no schema diff, data-only migration)        |

Addendum generated from the actual repository and database state — a direct pre/post query of every local organization's business-type data, plus a real browser-driven run of the onboarding wizard through Step 5 with a purpose-built 20-row CSV fixture — after this fix. All findings and test results were executed and observed, not projected.



## ADDENDUM

# Centro Implementation Report

## Content-Based Column Inference — Understanding Any Spreadsheet Structure

This addendum replaces header-alias matching as the client-import pipeline's primary mechanism with a genuine, content-based spreadsheet-analysis layer. The previous approach could only recognize a column it already had a name for; this one infers what every column means from the values themselves, works for CSV and XLSX identically, and asks a human to confirm only when the evidence is genuinely ambiguous.

|                   |                                                                                                          |
|-------------------|----------------------------------------------------------------------------------------------------------|
| Prepared for      | noammaarachot@gmail.com                                                                                  |
| Addendum covers   | The column-inference engine, its scoring model, the mapping-confirmation UI, and end-to-end verification |
| Applies on top of | M1–M13 report + Epic 2, Epic 3, Import Hardening & Business-Type Migration addenda                       |



## 18. Content-Based Column Inference

### 18.1 Problem

Every import pipeline up to this point — including the "hardened" one from the previous addendum — still located the client-name, phone, email, and business-type columns by matching header text against a fixed alias list ( `HEADER_ALIASES` in `src/lib/csv.ts` ). That works only for files whose headers happen to be one of the strings on the list. A real accounting office's export can rename, reorder, drop, or translate those headers, omit a header row entirely, or add unrelated columns — none of which a fixed alias list can anticipate. The requirement was explicit: Centro must understand a spreadsheet's structure from its data, not from having seen that exact header before, for *any* office's file, not a narrow fix for one test case.

### 18.2 Architecture

`src/lib/import/columnAnalyzer.ts` is a new, standalone module — no dependency on CSV or XLSX parsing — that takes the plain `string[][]` grid either format parser already produces and scores every column against every candidate role (name, phone, email, business type, notes) using three independent signal families, exactly as specified:

| Signal        | What it measures                                                                                                                                                                                                                                                                           | Max weight                               |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| Header text   | Exact or partial match against a known alias list, in Hebrew and English                                                                                                                                                                                                                   | 20–25% (never the deciding signal alone) |
| Value pattern | Fraction of non-empty cells matching the role's shape — an Israeli phone-number regex tolerant of separators/country-code/lost-leading-zero, an email regex, the existing Hebrew/English business-type synonym dictionary, or a lenient "looks like text, not a phone or email" name check | 40–75% (dominant signal for every role)  |
| Structure     | Cardinality relative to row count — a client-identity column (name/phone/email) should be close to one distinct value per row; a categorical column (business type) should repeat a small set of values across many rows                                                                   | 5–35%                                    |

The structure signal is what keeps a company-name column that happens to contain **כע"מ** from ever being mistaken for the business-type column — its values are still almost all unique, while a genuine business-type column repeats. Assignment is sequential and greedy — phone first (the least ambiguous pattern, a digit-string regex), then name, then business type (accepted only above its own confidence bar, otherwise left unmapped — requirement 9's "no business-type column found" path), then email, then notes — removing each claimed column before the next role is scored, so a column already confidently claimed for one role can never simultaneously mislead the confidence check for another.

**Confidence and the human fallback:** after every role is resolved, each required role's assigned column is checked against every column that ended up genuinely unclaimed by any role — if a close, unclaimed rival

exists, or the absolute score is weak, confidence is "low" and the wizard shows a short mapping-confirmation screen instead of guessing (requirement 8). An `inferColumnRolesViaAI` stub — the same documented "mock-first" pattern as `classifyViaAI` in the existing per-row classifier — sits between deterministic scoring and the human fallback, exactly matching the required priority order (deterministic first, AI second, human last), even though no LLM provider is configured for this pilot.

**Both formats, one pipeline:** `clientImportAdapter.ts` now exposes `parseClientImportFileToGrid()`, returning the same grid shape for CSV and XLSX; column understanding is implemented exactly once and used identically by both.

### 18.3 The Mapping-Confirmation Screen

When confidence is low, `importAndClassifyClients` (Step 4's action) returns a `needsMapping` payload instead of importing anything — the parsed grid, detected headers (synthesized as "N עמודה" when there is no header row), a sample of the data, and the analyzer's best guess per role. `Step4Import.tsx` renders this as a short form: one labeled dropdown per role ("לדעתנו זו עמודת שם הלקוח" / phone / email / business type), pre-filled with the suggestion, listing every column with a live sample value, plus a small preview table of the first rows. Submitting calls a new action, `confirmImportMapping`, which finishes the exact same import+classify logic (now factored into a shared `importParsedRows` helper) using the human-confirmed mapping — the original file never needs to be re-uploaded, since the parsed grid round-trips through the confirmation form as JSON.

### 18.4 Analysis Summary in the UI

Requirement 14: Step 5 now shows, above the existing imported/classified/needs-review banner, a row of badges naming exactly which column Centro used for each detected role and whether the match was confident or came from a manual confirmation. This is sourced from a new audit event, `clients.import_analyzed` (recorded by `importParsedRows` alongside the existing `clients.imported` / `clients.classified` events), read back by a new `getLatestAuditEventByType()` query — reusing the existing audit-log infrastructure rather than inventing a parallel persistence mechanism, since Step 5 is server-rendered after Step 4's redirect and has no other way to see that action's return value.

### 18.5 Row-Level Safety (Requirement 10)

Column-level detection and row-level classification are deliberately separate concerns. Once the business-type *column* is found — even from a mix of valid and noisy values (city names, free-text notes, stray numbers, blanks) — every individual row still goes through the existing, unchanged `classifyClientBusinessType()` (`src/lib/ai/businessTypeClassifier.ts`), which only ever assigns a type it can actually match and leaves anything else `null` (Unclassified, routed to Step 5's manual bulk-assignment flow). A noisy value degrading the column's detection score slightly never results in a wrong classification for that specific row — it results, correctly, in that one row needing manual review while every valid row around it still classifies automatically.

### 18.6 Automated Testing

`src/lib/import/columnAnalyzer.test.ts` — 18 new tests, covering every case listed in the requirement: detector-level tests for phone numbers across real-world formats (dashes, spaces, parens, international prefix, numeric-stored values with a lost leading zero) and for the full Hebrew/English business-type vocabulary; column-level tests for entirely generic/unknown headers, a missing header row, mixed valid and invalid business-type values (verifying noisy rows don't corrupt column detection while still passing through unclassified, never forced), company names instead of person names (verifying the structural cardinality signal keeps this from colliding with business-type detection), extra unrelated columns, ragged rows (a data row shorter than the header), duplicate phone numbers, and the low-confidence path itself (including a direct test of the manual-mapping build path the confirmation screen relies on). Combined with the existing suite, the project's total is now 46 passing tests across 5 files (up from 28/4).

`src/lib/csv.ts`'s `mapRowsToClientRows()` — now a thin convenience wrapper over the new analyzer — was verified to keep every one of its existing tests passing unchanged, along with the existing XLSX round-trip and adapter tests, so this is a genuine drop-in upgrade of the detection layer, not a breaking change to any existing contract.

18.7 End-to-End Verification

| Three real browser-driven scenarios, both file formats |                                                                        |                                                                                                                                                                           |
|--------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scenario                                               | File                                                                   | Result                                                                                                                                                                    |
| Clean, well-labeled file (regression check)            | CSV, standard headers                                                  | Unchanged behavior — straight to Step 5, high confidence, 6/6 classified, zero English leaked                                                                             |
| Entirely generic/unknown headers                       | XLSX, headers "פרטים"<br>"4 פרטים"-"1"                                 | Resolved automatically from content alone — high confidence, no confirmation screen, 11/11 imported and classified correctly                                              |
| Genuinely ambiguous structure                          | XLSX, two equally phone-shaped columns (mobile + fax), no header hints | Correctly triggered the mapping-confirmation screen listing both candidate columns; import completed successfully (4/4 classified) after confirming the suggested mapping |

All three scenarios were run as fresh registrations through the real onboarding wizard in a headless browser (not a unit-test simulation), confirming the analyzer, the confirmation UI, and the shared import+classify pipeline all work together correctly for both file formats.

18.8 Files Touched

| File                                          | Change                                          |
|-----------------------------------------------|-------------------------------------------------|
| <code>src/lib/import/columnAnalyzer.ts</code> | <b>new</b> — the content-based inference engine |

|                                                         |                                                                                                                                                                                                                                                    |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>src/lib/import/columnAnalyzer.test.ts</code>      | <b>new</b> — 18 tests                                                                                                                                                                                                                              |
| <code>src/lib/csv.ts</code>                             | <code>mapRowsToClientRows()</code> rebuilt on the analyzer; new <code>buildClientRowsFromMapping()</code> pure mapping-application helper                                                                                                          |
| <code>src/lib/import/clientImportAdapter.ts</code>      | new <code>parseClientImportFileToGrid()</code> raw-grid entry point shared by both formats                                                                                                                                                         |
| <code>src/lib/ai/businessTypeClassifier.ts</code>       | <code>matchSynonym</code> exported for reuse by the column analyzer                                                                                                                                                                                |
| <code>src/app/onboarding/actions.ts</code>              | <code>importAndClassifyClients</code> now analyzer-driven with a <code>needsMapping</code> result; new <code>confirmImportMapping</code> action; shared <code>importParsedRows</code> helper; new <code>clients.import_analyzed</code> audit event |
| <code>src/app/onboarding/steps/Step4Import.tsx</code>   | new mapping-confirmation sub-view with per-role dropdowns and a data preview                                                                                                                                                                       |
| <code>src/app/onboarding/steps/Step5Analysis.tsx</code> | new detected-columns/confidence summary banner                                                                                                                                                                                                     |
| <code>src/lib/data/auditLog.ts</code>                   | new <code>getLatestAuditEventByType()</code> query                                                                                                                                                                                                 |

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Addendum generated from the actual repository state — 46/46 automated tests, a clean production build and lint pass, and three real browser-driven end-to-end scenarios across both CSV and XLSX — after this change. All findings and test results were executed and observed, not projected.

## EPIC 4

# Centro Implementation Report

## The Smart Import Engine — A Document-Understanding Redesign of Client Import

A full architectural rebuild of client import, not an incremental fix: spreadsheet structure understanding, content-only column inference extended to eight field types, multi-layer business-type classification with numeric confidence and human-readable reasoning, and a per-organization learning mechanism — so Centro adapts to any real accounting office's file, never the other way around.

|                   |                                                                                                                                                 |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Prepared for      | noammaarachot@gmail.com                                                                                                                         |
| Addendum covers   | Spreadsheet understanding, column inference, the classification pipeline, confidence scoring, the learning mechanism, and why the design scales |
| Applies on top of | M1–M13 report + Epic 2, Epic 3, Import Hardening, Business-Type Migration & Column-Inference addenda                                            |

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## 19. The Smart Import Engine (Epic 4)

### 19.1 Why a Redesign, Not a Fix

The previous addendum's column-inference layer solved "don't trust the header text" for four fields. It did not solve everything a real accounting office's export actually looks like: a title row above the table, a "total" row below it, several worksheets in one workbook, columns in an arbitrary order, fields Centro had never modeled (city, company registration number, tax ID), business-type values missing outright, or an office's own shorthand no dictionary could know in advance. This epic treats import as a document- understanding problem end to end — from "what is this file's actual structure" through "what does each column mean" to "what does this specific office call this business type" — rather than patching the previous column-matcher one gap at a time.

NEW MODULES

**2**

(**spreadsheetStructure**,  
**+xlsxParser rewrite**)

COLUMN ROLES  
RECOGNIZED

**8 (was 5)**

NEW DB OBJECTS

**1 table, 3 columns**

CLASSIFICATION LAYERS

**4 (learned →  
dictionary →  
context → AI)**

AUTOMATED TESTS

**67 (+21 this epic)**

### 19.2 STEP 1 — Spreadsheet Understanding

Before any column is scored, the file is understood as a document, not just a grid:

- **Multiple worksheets** ( `xlsxParser.ts` 's new `parseXlsxWorkbook` ) — every sheet is read, and the one with the most non-empty rows is used as the client table, so a workbook with a cover/instructions sheet never accidentally imports the wrong one. Every sheet name found, and which was used, is reported in the wizard's understanding summary.
- **Merged cells** — verified directly against ExcelJS's actual behavior rather than assumed: a merged range's value is already resolved onto every cell in that range by the time the workbook finishes loading (confirmed with a throwaway script before writing any handling code), so a business-type label merged across several client rows is never lost. The engine detects and reports that merges were present; no extra propagation code was needed once that was verified.
- **Hidden columns** — detected via each column's `hidden` flag and reported, but never excluded: a hidden column is frequently still real data an office just collapsed for display.
- **Where the table actually starts and ends** ( `spreadsheetStructure.ts` , format-agnostic — the same function trims a CSV or an XLSX grid) — the table's real column count is estimated as the most common non-empty-cell-count across all rows; a leading title row or trailing "total: N clients" row won't share that width (or matches a small set of summary keywords) and is trimmed before column analysis ever sees it.

### 19.3 STEP 2 — Column Detection, Extended



`columnAnalyzer.ts` (from the previous addendum) now scores eight roles instead of five — name, phone, email, business type, and notes as before, plus city, company ID (פ.ח), and tax ID. The three new roles are deliberately **best-effort, never required, never gating confidence**: city is a categorical value matched against a curated Israeli city list; company/tax IDs are structurally identical (both plain 9-digit numbers) and, honestly, cannot be told apart — or fully told apart from a phone number missing its leading zero — from content alone, so header text is weighted far more heavily for these two roles specifically than for any other. That limitation is documented in code rather than glossed over.

#### A real bug found during verification: shuffled-column tie collision

Testing with a genuinely shuffled, header-less file surfaced a bug the previous addendum's fixtures never exposed: a business-type column made of varied synonym forms ("עמותה", "פסור", "מורשה", "מ", "חברה בע"מ", "Payroll") has five *literally distinct* string values, so it can score identically to a genuine name column on the "name" role (same pattern and cardinality signals). The previous tie-break — whichever column appears first — happened to always pick correctly by coincidence of column order in every earlier test file; a shuffled file with the business-type column positioned before the name column exposed the real flaw: it stole the name role outright.

**Fix:** ties are now broken by *specificity* — a column's score for a role minus its own best score among every other role. The real name column has a wide gap between "name" and its next-best role; the colliding business-type column doesn't (it also scores well on "business type" itself). This is a genuine algorithmic improvement, not a special case for the one file that exposed it — a regression test using that exact shuffled shape is now part of the suite.

### 19.4 STEP 3 — Multi-Layer Business-Type Classification

`classifyClientBusinessType()` (`src/lib/ai/businessTypeClassifier.ts`) now runs a strict four-layer priority chain per client row, never a single cell in isolation:

1. **Learned synonyms** (see 19.6) — this specific organization's own past correction, checked first because it is the most specific signal available.
2. **Global deterministic dictionary** on the explicit business-type column, when present — expanded this epic to cover every documented shorthand (ע. מורשה, ע. פסור, מלכ"ר, חל"צ) alongside the existing full forms.
3. **Context inference** — when the explicit column is empty or unrecognized, the same two dictionaries are checked again against the client's name, then every other non-empty cell in the row (city, notes, an unmapped extra column — everything `columnAnalyzer.ts`'s new `extractRowContextValues()` collects). A company named "אלפא פתרונות בע"מ" with a blank type column still classifies correctly from its own name; this is still deterministic matching, just against different input, never a guess.
4. **AI fallback** — reached only once every deterministic layer has failed, given the *entire row*, not one cell, and contracted to return a business type with its own confidence and a reason. No LLM provider is configured for this pilot, so this remains a documented no-op stub (the same "mock-first" pattern as every other AI seam in this codebase) — never a fake guess, and wiring one in later touches only this function.

Every result carries a human-readable `reason` string alongside its confidence, for the audit trail and future UI surfacing.

## 19.5 STEP 4 — Confidence Scoring

| Band    | Behavior                                                            | Typical source                                                                      |
|---------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 95–100% | Auto-classified silently                                            | Explicit column, exact/synonym dictionary match, or a learned synonym (all 99%)     |
| 70–94%  | Auto-classified, flagged "suggested" in the UI                      | Context inference (85%) — the client's name or another cell, not a dedicated column |
| <70%    | Never applied — left Unclassified, routed to manual bulk assignment | No deterministic or AI signal found anywhere in the row                             |

`clients.businessTypeConfidence` (new column) persists whichever score applied, so Step 5 can distinguish "certain" from "suggested" per client, not just aggregate counts.

## 19.6 STEP 7 — Learning From Corrections

### Scoped strictly per organization — verified with two separate import sessions

A new `business_type_learned_synonyms` table (organization ID, business type ID, normalized source text, unique per org+text) is written the moment a human manually assigns a Business Type to a client whose import row carried unrecognized raw text ( `clients.importedBusinessTypeText` , also new). Every subsequent import for that same organization checks this table first — before the global dictionary — so the office's own shorthand is recognized immediately, with no global retraining and zero effect on any other organization's data.

Verified end to end, not just unit-tested: a client whose business-type column read the made-up term "עוסק-מ" (deliberately absent from any dictionary) imported as Unclassified in one browser session; manually assigning it to עוסק מורשה in Step 5 recorded the synonym; a second, independent browser session — a fresh login, not a continuation — then imported two new clients using the exact same term and both classified automatically, with zero manual review needed.

## 19.7 STEP 6 — Stable Internal Model

This epic closes out a real incident from two addenda ago: renaming the five starter Business Types' display label from English to Hebrew silently broke every organization's classification and UI, because matching depended on the exact display string. `business_types` now has a `canonical_key` column — a stable, renaming-proof identity ( `limited_company` , `authorized_dealer` , `exempt_dealer` , `nonprofit` , `payroll_only` ) for the five standard types, backfilled onto every existing organization's rows in the same migration. The classifier now matches a synonym to a canonical key, then looks up the organization's row by that key — `business_types.name` is purely a display label from here on, freely editable per-organization exactly as before, with no way for a rename to ever again silently break matching. Org-created custom types have no global canonical concept and are still matched by name, unaffected.

## 19.8 Why This Scales

- **Deterministic-first, always.** Every layer — column detection, business-type matching, learned synonyms — is content-based pattern matching against explicit, reviewable dictionaries, not a model call. The (currently unconfigured) AI fallback is structurally the last resort in both the column-detection and classification pipelines, never the first thing tried, so accuracy for the overwhelming majority of real files doesn't depend on an LLM being available, fast, or cheap.
- **No per-office code.** Nothing in this epic references a specific file, header, or office. Every fixture used for verification was deliberately constructed to be structurally different from the others (shuffled columns, no headers, multiple sheets, title/footer rows, missing values) specifically to prove the engine generalizes rather than having been tuned to one example.
- **Learning is additive and isolated.** Each organization's learned synonyms only ever improve that organization's own future imports; there is no shared model, no cross-tenant leakage, and no retraining step — a new row in one small table is the entire mechanism.
- **Confidence gates automation, not accuracy claims.** The 95/70 bands mean a low-signal row is always routed to a human rather than silently guessed — the system's accuracy on real files improves the false-negative rate (fewer wrongly-Unclassified rows), never the false-positive rate (a wrongly-classified row), which is the property that actually matters for a system managing real client data.

## 19.9 Verification

### 67 automated tests (21 new) + 3 real browser-driven scenarios

New/extended suites: `columnAnalyzer.test.ts` (bonus roles, row- context extraction, the shuffled-column regression), `spreadsheetStructure.test.ts` (new — table-bounds detection), `xlsxParser.test.ts` (multi-sheet/ merge/hidden-column metadata), `clientImportAdapter.test.ts` (STEP 1 structure detection end to end), and a full rewrite of `businessTypeClassifier.test.ts` for the new row-context signature, canonical keys, confidence bands, and learned synonyms.

| Scenario                                                                                                         | Covers                                                                                 | Result                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CSV with a title row, a footer summary row, missing business-type values, and one not-yet-learned shorthand term | STEP 1 table bounds, STEP 3 context inference, confidence tiers, the unclassified path | 6 imported (title/blank/footer correctly excluded), 3 auto-classified (99%), 2 suggested via context inference (85%), 1 correctly left for manual review                       |
| XLSX with a cover sheet + data sheet, no header row, shuffled column order, plus city and company-ID columns     | STEP 1 multi-sheet selection, STEP 2 header-less + shuffled detection, bonus roles     | Correct sheet selected automatically and reported; all 6 roles (including city and company ID) detected correctly despite the shuffle; 5/5 classified at 99%                   |
| Two independent browser sessions against the same organization, days apart in real usage terms                   | STEP 7 learning mechanism, org-scoping, persistence                                    | A manually corrected term was recognized automatically on the next import with zero manual review — proven across a genuinely separate login session, not just in-memory state |

## 19.10 Files Touched

| File                                                                                      | Change                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>src/db/schema.ts</code>                                                             | + <code>business_types.canonical_key</code> ,<br>+ <code>clients.businessTypeConfidence / importedBusinessTypeText</code> ,<br>new <code>business_type_learned_synonyms</code> table |
| <code>drizzle/0010_tense_puma.sql</code>                                                  | <b>new</b> — schema + canonical-key backfill migration                                                                                                                               |
| <code>src/lib/import/spreadsheetStructure.ts</code>                                       | <b>new</b> — format-agnostic table-bounds detection                                                                                                                                  |
| <code>src/lib/import/xlsxParser.ts</code>                                                 | rebuilt around <code>parseXlsxWorkbook()</code> — multi-sheet selection,<br>merge/hidden-column reporting                                                                            |
| <code>src/lib/import/columnAnalyzer.ts</code>                                             | +city/companyId/taxId roles, row-context extraction, specificity tie-break fix                                                                                                       |
| <code>src/lib/import/clientImportAdapter.ts</code>                                        | new <code>analyzeImportFileStructure()</code> STEP-1 entry point                                                                                                                     |
| <code>src/lib/csv.ts</code>                                                               | <code>CsvClientRow.otherValues</code> for row-context classification                                                                                                                 |
| <code>src/lib/ai/businessTypeClassifier.ts</code>                                         | rebuilt — canonical-key matching, expanded synonyms, 4-layer pipeline, 0-100 confidence, reasons                                                                                     |
| <code>src/lib/businessTypes.ts</code>                                                     | canonical keys on starters, learned-synonym read/write, learning hook in <code>assignClientsToBusinessType</code>                                                                    |
| <code>src/app/onboarding/actions.ts</code>                                                | full import-pipeline rewrite around the new structure/classification layers                                                                                                          |
| <code>src/app/onboarding/steps/Step4Import.tsx</code><br>/ <code>Step5Analysis.tsx</code> | structure metadata round-trip; "Centro understood your file" checkmark summary with confidence tiers                                                                                 |
| 6 test files (2 new)                                                                      | 67 tests total (+21)                                                                                                                                                                 |

Addendum generated from the actual repository state — 67/67 automated tests, a clean production build and lint pass, and three real browser-driven end-to-end scenarios spanning CSV and XLSX, headers and no headers, ordered and shuffled columns, and two independent login sessions proving the learning mechanism — after this redesign. All findings and test results were executed and observed, not projected, including the shuffled-column bug found and fixed during this verification pass.

## EPIC 5 — KICKOFF

# Centro Implementation Report

## Architecture Alignment Roadmap & Milestone 1: Manual-Entry Classification Parity

A new phase begins: aligning the codebase against a formal Architecture & Philosophy document for the first time. This addendum covers the planning process that produced the 7-milestone roadmap, an architectural correction made during planning itself, and the first completed milestone.

|                   |                                                                                            |
|-------------------|--------------------------------------------------------------------------------------------|
| Prepared for      | noammaarachot@gmail.com                                                                    |
| Addendum covers   | Architecture review process, roadmap approval, Milestone 1 implementation and verification |
| Applies on top of | M1–M13 report + Epic 2, 3, 4 addenda                                                       |



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## 20. Architecture Alignment Roadmap — Planning & Milestone 1

### 20.1 Context

The user supplied a formal *Centro Architecture & Philosophy* document (8 chapters, since revised to a live in-repo Markdown source — `ARCHITECTURE.md`) as the project's architectural source of truth, and requested a planning-only phase before any code changed: read the document, compare it against the real implementation, and produce a full gap analysis and milestone roadmap.

### 20.2 Architecture Review & Gap Analysis (summary)

The review was done against the actual codebase — full schema, the collection/ conversation state machine, the scheduler, and all three "AI" classifiers — not from memory. Findings:

- **Already strong:** the Epic 4 business-type classifier is a near-complete reference implementation of the philosophy's Learning/Decision- Hierarchy chapters for one narrow domain — a template to generalize, not rebuild.
- **Partial:** document classification has confidence-gated auto-approval but no "Learned Knowledge" tier; intent classification is logged but never acts; the audit trail is comprehensive but has no severity/exceptions concept beyond the documents queue.
- **Missing entirely:** no "Learning Mode" state on a client; no post-cycle analysis; no client-facing WhatsApp confirmation mechanism (the one working "learn from a correction" example — business-type synonyms — is confirmed by the accountant, never the client); learning is siloed to business-type classification only; clients created outside the onboarding wizard ( `/clients/new` ) received no classification at all — confirmed by direct inspection of `clients/actions.ts` , which had zero business-type handling before this milestone.

One conflict was surfaced rather than resolved silently: the document's stated Decision Hierarchy order (Business Rules → Learned Knowledge → AI → Human Review) is the reverse of what the working business-type classifier actually does (Learned Knowledge checked first). **Resolved by user decision: keep the implementation, update the document** — Chapter 6 of `ARCHITECTURE.md` now states the hierarchy as actually implemented, with the reasoning documented directly in the chapter (§20.6 below has the exact wording).

### 20.3 An Architectural Correction Made During Planning, Not After

#### The original Milestone 6 draft violated the product's own boundaries

The first roadmap draft's Milestone 6 ("Learned Reminder-Timing Suggestions") would have had Centro infer personalized reminder cadence from observed client behavior. The user corrected this immediately and sharply: **Centro is a document collection platform — it learns only which documents to collect, never office policy, timing, or the client's business.** Business hours, working days, reminder cadence, and collection start day are accountant-configured, permanently, with no automated adaptation.

### Corrected: Milestone 6 became "Adaptive Document Collection"

Replaced with a milestone that learns exactly one thing — a client's required document list — through the same Observe → Suggest → Confirm → Learn lifecycle, with careful wording discipline added on a second pass (per further user correction): Centro never states a document "is no longer required," only that it *appears* to no longer be part of the regular collection, and always asks before ever removing it from a client's profile.

This correction also revealed that **Milestone 2** (originally "Learning Mode & First-Cycle Analysis") had been scoped to capture cycle-duration and reminder-count facts — timing-adjacent data whose only purpose was feeding the now-removed timing-personalization milestone. Milestone 2 was trimmed accordingly, down to exactly the Learning Mode flag itself, with no timing capture at all — avoiding a component that would have had no legitimate remaining consumer.

The resulting permanent principle — **"Centro learns only which documents should be requested from each client. Centro never learns or modifies office policies such as business hours, working days, reminder cadence, or collection dates."** — is now Chapter 8 of `ARCHITECTURE.md`, explicitly scoped as a constraint on every current and future milestone, not only Milestone 6.

## 20.4 The Approved Roadmap

| # | Milestone                               | Status                                                                                                                                                                                                                          |
|---|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Manual-Entry Classification Parity      | <b>Complete (this addendum)</b>                                                                                                                                                                                                 |
| 2 | Learning Mode (client-level flag only)  | Planned                                                                                                                                                                                                                         |
| 3 | Document Classification Learning        | Planned                                                                                                                                                                                                                         |
| 4 | Unified Exceptions & Review Surface     | Planned                                                                                                                                                                                                                         |
| 5 | Client-Facing Confirm-via-WhatsApp Loop | Planned                                                                                                                                                                                                                         |
| 6 | Adaptive Document Collection            | Planned                                                                                                                                                                                                                         |
| 7 | Pilot Hardening                         | Planned — final milestone: end-to-end regression across M1–M6, edge-case and org-isolation audit, realistic-data-volume performance check, security review, documentation consolidation, dead-code cleanup, full-app smoke test |

## 20.5 Documentation Process, Formalized



Per explicit user decision: `ARCHITECTURE.md` is now the version-controlled source of truth in the repository (joining `AGENTS.md` / `CLAUDE.md`), rendered to `reports/Centro-Architecture-and-Philosophy.pdf` after every milestone. The Implementation Report continues exactly as before — an append-only PDF history, one addendum per milestone.

### 20.6 Milestone 1 — Manual-Entry Classification Parity

**Goal:** a client created through `/clients/new` (not the bulk-import wizard) now goes through the exact same classifier, with the exact same organization-scoped learned synonyms and confidence bands, as an imported client — closing the gap found in §20.2.

**What changed:**

- `createClient` (`clients/actions.ts`) now runs `classifyClientBusinessType()` against the typed name via context inference (no explicit business-type field exists on this form — the same path a CSV row with an empty business-type column already takes). Applied automatically at  $\geq 70\%$  confidence, exactly like import; left Unclassified below that, never guessed.
- The confidence-band constants (95 / 70) were promoted from a local, duplicated definition inside the onboarding wizard's action file into `businessTypeClassifier.ts` itself and exported — a single source of truth now shared by both the wizard and this new call site.
- New `setClientBusinessType` action plus a "Business Type" card on the client detail page (`/clients/[id]`) — shows the current type with a certain/suggested badge, or an honest "Centro couldn't determine this automatically" message, with a picker (existing type or a new one) always available. A manual choice is stored at 100% confidence — a human decision is definitionally certain, not a guess.

**Two issues found and fixed during verification (not left for later)**

**Test-script false positive:** the verification script initially expected a clear legal-suffix match (e.g. **אבגד ח פתרונות בע"מ**) to show as "certain" (95%+). It correctly showed "suggested" (85%) instead — by design, context-inference (matching from a name alone, not an explicit column) is always scored 85, never 95+, which is reserved for explicit-column or learned- synonym matches. The test's expectation was wrong, not the app; fixed in the script.

**Real, small UX bug:** the new "Business Type" card's assignment button and the pre-existing "assign service" button on the same page were both labeled simply **"שיוך"** — ambiguous for automated testing and mildly ambiguous for a human skimming the page too. Renamed to **"שיוך סוג עסק"** for clarity.

### 20.7 Verification

| Scenario                                                                           | Result                                                                                |
|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Manually create a client with a clearly classifiable name (company-suffix pattern) | Classified correctly, shown as "suggested" (85%) — never falsely presented as certain |

|                                                             |                                                                                        |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Manually create a client with no classifiable signal at all | Left honestly Unclassified — "Centro couldn't determine this automatically" — no guess |
| Manually assign a business type from the detail page        | Stored at 100% confidence, shown as certain, badge updates immediately                 |

All three scenarios were run as a real, fresh registration through the full onboarding wizard into the live app shell in a headless browser — not simulated. 67/67 existing automated tests continue to pass unchanged (no new pure logic was introduced; this milestone wires already-tested classifier logic into a new call site). Clean production build and lint.

## 20.8 Files Touched

| File                                                        | Change                                                                                  |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <code>ARCHITECTURE.md</code>                                | <b>new</b> — version-controlled source of truth                                         |
| <code>reports/Centro-Architecture-and-Philosophy.pdf</code> | <b>new</b> — rendered PDF, regenerated every milestone                                  |
| <code>src/lib/ai/businessTypeClassifier.ts</code>           | exported<br><code>AUTO_CLASSIFY_CONFIDENCE</code> / <code>SUGGESTED_CONFIDENCE</code>   |
| <code>src/app/onboarding/actions.ts</code>                  | now imports the shared confidence constants instead of a local duplicate                |
| <code>src/app/(app)/clients/actions.ts</code>               | <code>createClient</code> now classifies; new <code>setClientBusinessType</code> action |
| <code>src/app/(app)/clients/[id]/page.tsx</code>            | new "Business Type" card with confidence badge and manual override                      |

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Addendum generated from the actual repository state — a real browser-driven verification pass, 67/67 automated tests, and a clean build/lint — after this milestone. All findings were executed and observed, not projected.

# Learning Mode

Chapters 1 and 2 of the Architecture & Philosophy document state that every client begins in Learning Mode. This milestone makes that a real, queryable fact rather than prose.

## 21. Learning Mode

### 21.1 What Was Built

`clients.learningMode` (boolean, default `true`) and `clients.firstCycleCompletedAt` (nullable timestamp) — two new columns, additive migration. `collectionRequestStateMachine.ts`'s `applyTransition()` now calls a small `exitLearningModeIfFirstCycle()` helper whenever a transition successfully reaches `completed`: it flips the flag and stamps the timestamp, scoped by a `WHERE learning_mode = true` clause so it is naturally idempotent — a client's second and every later completed cycle is a silent no-op there, with no extra guard logic needed.

Deliberately excluded, per the Chapter 8 boundary established during roadmap planning: no cycle-duration, reminder-count, or any timing data is captured. This milestone's only purpose is gating Milestone 6's "document appears to no longer be needed" detection — a client with zero completed cycles has no pattern to compare against, so that check must never fire for them.

A small "מצב למידה" (Learning Mode) badge was added to the client detail page header — visible, quiet, informational, consistent with Chapter 7's "invisible but not hidden" automation: an accountant can see the state without it ever demanding attention.

### 21.2 Verification

#### A real bug hunt, resolved as a test-script artifact, not an app bug

The first verification pass appeared to fail: after completing a client's first full collection cycle end to end (draft → active → processing → completed, with both starter requirements satisfied), the badge still showed on re-navigation. A direct database query — run standalone, after stopping the dev server per this project's established PGlite single-process discipline — confirmed `learning_mode = false` and `first_cycle_completed_at` correctly set. The failure was Next.js's client-side router cache serving a stale RSC payload for a URL the same browser tab had already visited earlier in the run; re-checking with a fresh page (no navigation history) confirmed correct behavior immediately. Root-caused before concluding, not assumed.

Full scenario, run live in a headless browser: registered a fresh organization, created a client (confirmed Learning Mode badge present), manually assigned it **עוסק פטור** (2 starter requirements), opened a Collection Request, satisfied both requirements, and drove the request through to `completed`. Confirmed the badge was

gone on the next fresh page load. Clean build, lint, and 67/67 existing tests (no new pure logic — this milestone is entirely a state-machine hook, verified live).

21.3 Files Touched

| File                                     | Change                                                         |
|------------------------------------------|----------------------------------------------------------------|
| src/db/schema.ts                         | + clients.learningMode ,<br>+ clients.firstCycleCompletedAt    |
| drizzle/0011_lyrical_shiver_man.sql      | <b>new</b> — additive migration                                |
| src/lib/collectionRequestStateMachine.ts | new exitLearningModeIfFirstCycle() hook on the completion path |
| src/app/(app)/clients/[id]/page.tsx      | Learning Mode badge in the page header                         |

Addendum generated from the actual repository and database state — a real browser- driven end-to-end collection cycle, a direct post-hoc database verification, and a clean build/lint/test pass — after this milestone.

# Document Classification Learning

Generalizes Chapter 6's Decision Hierarchy to a second real domain: document classification now learns from a client's own manually-corrected history, not just business-type synonyms.

## 22. Document Classification Learning

### 22.1 What Was Built

New `document_learned_patterns` table (organization + client scoped, one row per manual correction — filenames vary too much month to month for the exact-text lookup business-type synonyms use, so this is fuzzy token-overlap matching against a client's own history instead). Written from exactly one place: `assignDocumentRequirement` — whenever an employee manually assigns an unmatched document to a requirement, the requirement's stable template identity ( `serviceDocumentRequirements.id` , not the frozen per-cycle snapshot row) and the filename are recorded.

`documentClassifier.ts` gains a full 4-layer pipeline — `classifyDocumentWithLearning()` — mirroring `businessTypeClassifier.ts` 's structure: this client's learned patterns first, then the existing deterministic filename-vs-requirement-name heuristic (now formally "the Business Rules layer"), then a new documented AI-fallback stub (no-op, no provider configured — same mock-first pattern as every other AI seam in this codebase), falling through to `needs_review` (Human Review) exactly as before. The plain heuristic remains separately exported and unchanged in behavior for any caller with no learning history to check.

### 22.2 Verification

12 new unit tests, including a direct proof that the with-learning pipeline produces byte-identical results to the plain heuristic when no learning exists yet (no behavior change for any client until their first correction). 81/81 tests total.

#### A pre-existing discrepancy found, not introduced — logged for Milestone 7, not fixed here

Writing the first-ever test file for this module surfaced that `isFuzzyDuplicate` 's own doc comment example ("`bank-statement.pdf`" vs "`bank-statement-copy-2.pdf`") does not actually clear its own 0.7 token-overlap threshold (it scores 0.5) — a documentation/behavior mismatch that predates this milestone and was never unit-tested before. The logic itself is untouched by this milestone and the direction of the error is the safe one (slightly stricter than intended, so fewer false "duplicate" flags, not more) — logged as a candidate for Milestone 7's edge-case audit rather than changed here, to keep this milestone's diff scoped to what it actually set out to do.

### End-to-end proof of the full Observe → Suggest → Confirm → Learn loop

Run live in a headless browser against a fresh organization: uploaded "pdf.ינואר\_102\_טופס" (a filename sharing no tokens with either of the client's two requirement names) — correctly landed in the unmatched- documents queue (Observe/Suggest: the system honestly couldn't match it). An employee manually assigned it (Confirm). A second, later upload for the *same client* — "pdf.פברואר\_102\_טופס", still sharing no tokens with either requirement's name — was **auto-classified and auto-approved** purely from the one learned pattern (Learn, then applied), with zero manual review required the second time.

## 22.3 Files Touched

| File                                             | Change                                                                                                             |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| src/db/schema.ts                                 | new document_learned_patterns table                                                                                |
| drizzle/0012_small_titanium_man.sql              | <b>new</b> — additive migration                                                                                    |
| src/lib/ai/documentClassifier.ts                 | rebuilt — <code>classifyDocumentWithLearning()</code> 4-layer pipeline, learned-pattern matching, AI-fallback stub |
| src/lib/ai/documentClassifier.test.ts            | <b>new</b> — 12 tests, first-ever coverage of this module                                                          |
| src/lib/documentLearning.ts                      | <b>new</b> — read/write seam for a client's document-learning history                                              |
| src/app/(app)/collections/conversationActions.ts | inbound document classification now goes through the learning pipeline                                             |
| src/app/(app)/collections/actions.ts             | <code>assignDocumentRequirement</code> now records the learned pattern                                             |

Addendum generated from the actual repository state — 81/81 automated tests, a clean build/lint pass, and a real browser-driven end-to-end proof of the learning loop — after this milestone.

# Unified Exceptions & Review Surface

A single, discoverable place for everything Centro wants an accountant's attention on — extending the dashboard's existing queue pattern rather than building a new, separate screen.

## 23. Unified Exceptions & Review Surface

### 23.1 What Was Built

A sixth dashboard KPI card, "הצעות סיווג לקוחות" (Client Classification Suggestions), sitting beside the pre-existing "דורש בדיקת עובד" (documents needing review) card. Its count and drill-down are clients whose business-type confidence is non-null and below the 95% "certain" threshold — i.e. exactly the "suggested" band established in Epic 4 — never the ones Centro is already sure about.

Implemented as its own query and its own drill-down branch rather than forced through the existing collection-request-shaped `QueueRow / listQueue()` machinery: a client-level exception (which business type) and a collection-request-level one (missing documents, stale conversations) are genuinely different shapes of data, and force-fitting one into the other's table columns would have produced a confusing, mislabeled table rather than a clearer one. Both now sit on the same dashboard, discoverable the same way, without pretending they're the same kind of row.

### 23.2 Verification

Run live: created one client with a name that only resolves to a "suggested" (85%) classification and one with no classifiable signal at all. Confirmed the new card appears on the dashboard, its drill-down lists only the suggested client (never the unclassified one — an unclassified client isn't a suggestion, it's an open item already visible on the client's own page), shows the correct type and confidence badge, and that clicking through lands on the correct client. Clean build, lint, and 81/81 existing tests (no new pure logic — this milestone is query composition and UI).

### 23.3 Files Touched

| File                                          | Change                                                                                                  |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <code>src/lib/data/dashboard.ts</code>        | new <code>business_type_suggestions</code> queue, count, and <code>listBusinessTypeSuggestions()</code> |
| <code>src/app/(app)/dashboard/page.tsx</code> | new KPI card and its own client-shaped drill-down table                                                 |

Addendum generated from the actual repository state — a real browser-driven verification pass, 81/81 automated tests, and a clean build/lint — after this milestone.

## Client-Facing Confirm-via-WhatsApp Loop

The one step of Chapter 3's Observe → Suggest → Confirm → Learn lifecycle that didn't exist anywhere in the codebase yet: Centro asking the client, not just the accountant, a short question — and acting on the reply.

### 24. Client-Facing Confirm-via-WhatsApp Loop

#### 24.1 What Was Built

A new domain-agnostic `pending_confirmations` table — a `kind` + free-form `payload` let any future caller attach its own meaning without this table knowing anything about documents, business types, or anything else. `src/lib/pendingConfirmations.ts` is the whole seam: `createPendingConfirmation()` sends the question over the (still-mocked, project-wide) WhatsApp transport and records exactly one row per question — Chapter 4's "asked only once" is structural here, not a policy someone has to remember. Replies resolve two ways, both real: a free-text client reply is parsed deterministically (Hebrew/English yes/no keyword matching, same mock-first pattern as every other classifier in this codebase) via `resolveConfirmationFromReply()`; an employee can also mark one resolved directly, the same quick-action pattern the existing `markFinished` / `markMoreDocuments` buttons already established.

**Not throwaway infrastructure** — this milestone ships a real, useful trigger point, not just a library nobody calls yet: on the collection request page's "unmatched documents" assignment form, an employee can now optionally ask the client "should we request this document every month?" at the exact moment of manual correction. Milestone 6 will add the automatic version of this (detecting recurrence without a human having to notice it first), reusing this exact mechanism.

Extends Milestone 4's unified exceptions surface with a fourth queue, "מחתיך לאישור לקוח" — every organization-wide still-open confirmation, one click from the dashboard.

#### 24.2 A Real Bug Found and Fixed During Verification

##### Two unrelated files matched purely because they shared a file extension

Milestone 5's own end-to-end test uploaded a second, genuinely unrelated document for the same client right after teaching Centro its first correction. It auto-classified — incorrectly. Root cause:

`jaccardTokenSimilarity()` (the token-overlap comparison Milestone 3 introduced) compared full filenames including their extension. Since nearly every document in this system is a PDF, two files sharing nothing but "**pdf.**" already shared one "free" token — enough, in this case, to cross the match threshold for two files that shared no meaningful words at all.



### Fixed at the source, not worked around

`jaccardTokenSimilarity()` now strips the extension before comparing, for both of its callers (`matchLearnedPattern` and `isFuzzyDuplicate`) — exactly matching what `classifyDocument()`'s own heuristic was already doing correctly. A regression test locks in the exact scenario that exposed it. Every existing test was re-verified by hand against the fix before it landed; none of them depended on the old, incorrect behavior (extension-token inflation happened to not change any existing test's pass/fail outcome, just the precision of the underlying number) — 87/87 passing, one new.

## 24.3 Verification

Full round trip run live in a headless browser: an employee manually assigned an unmatched document and checked "ask the client" — confirmed the WhatsApp question was actually sent and a pending-confirmation card appeared on the collection page and the dashboard. Resolved it via a simulated client reply of "כן" — confirmed automatic resolution and disappearance from both queues. Ran a second, independent confirmation through to resolution via the employee's manual "הלקוח סירב" override button instead, proving both resolution paths work. 87/87 tests, clean build/lint.

## 24.4 Files Touched

| File                                                                     | Change                                                                                |
|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <code>src/db/schema.ts</code>                                            | new <code>pending_confirmations</code> table                                          |
| <code>drizzle/0013_glorious_zodiak.sql</code>                            | <b>new</b> — additive migration                                                       |
| <code>src/lib/pendingConfirmations.ts</code>                             | <b>new</b> — domain-agnostic confirmation infrastructure + reply parsing              |
| <code>src/lib/pendingConfirmations.test.ts</code>                        | <b>new</b> — 5 tests, including the "don't misread an uncertain sentence" fix         |
| <code>src/lib/ai/documentClassifier.ts</code>                            | bug fix — extension-stripped filename comparison                                      |
| <code>src/app/(app)/collections/actions.ts</code>                        | optional "ask the client" on manual document assignment                               |
| <code>src/app/(app)/collections/conversationActions.ts</code>            | reply-based resolution; new <code>respondToConfirmation</code> manual-override action |
| <code>src/app/(app)/collections/[id]/page.tsx</code>                     | pending-confirmation card with quick-action buttons                                   |
| <code>src/lib/data/dashboard.ts</code> / <code>dashboard/page.tsx</code> | fourth unified-exceptions queue                                                       |

Addendum generated from the actual repository state — a real browser-driven end-to-end round trip through both resolution paths, 87/87 automated tests, and a clean build/lint — after this milestone, including the bug found and fixed during its own verification.



# הקובץ לא נמצא

ייתכן שהוא הועבר, נערך או נמחק.

ERR\_FILE\_NOT\_FOUND



# הקובץ לא נמצא

ייתכן שהוא הועבר, נערך או נמחק.

ERR\_FILE\_NOT\_FOUND



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ERR\_FILE\_NOT\_FOUND

# 25. Landing Page — Redesign, Trial CTA, and Mobile UX Fixes

|               |                                                                                                                                                                                                                                                                |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope         | Public marketing landing page only ( <code>src/components/landing/*</code> , <code>src/app/page.tsx</code> , <code>src/app/layout.tsx</code> metadata). No application/authenticated-app code touched.                                                         |
| Commits       | 3 — <code>b37af20</code> , <code>07fdfe6</code> , <code>747bc3a</code>                                                                                                                                                                                         |
| Files changed | 21 files, +589 / -194 lines                                                                                                                                                                                                                                    |
| Verification  | tsc, ESLint, full Vitest suite (138 tests), production build — clean on every commit; every fix additionally live-tested with Playwright against real 360/375/390/414/430px mobile viewports and desktop, plus a final pass against the live production domain |

## 25.1 Summary

The landing page was rebuilt across three rounds of work: a full copy and positioning rewrite (multi-industry messaging, a recurring trial-signup call to action, removal of "pilot" framing and of the Gmail integration mention), a first pass of mobile-specific bug fixes, and a final round of corrections after live re-testing surfaced two of the first-round fixes as incomplete. This chapter documents the whole arc as it stands at the end of the third round, including two real bugs found and fixed mid-implementation that were not part of the original request.

## 25.2 Trial CTA and Copy Rewrite

A new reusable component, `TrialCta.tsx` , renders a consistent "14-day free trial" call to action — always the site's existing purple-to-blue button gradient, never green (green stays reserved for WhatsApp everywhere on the site) — with the required reassurance copy ("no credit card, no commitment") beneath the headline. It appears in four places: the Hero as the primary action, the mid-page trust card that previously pitched joining a pilot, the bottom `FinalCTASection` , and a short nudge shown after the interactive demo finishes playing.

Landing copy was rewritten to speak to any business that collects documents from clients, not specifically accounting firms: the Hero gained an explicit "suitable for any business that collects documents from clients" line, testimonial roles were diversified (a dental clinic, an insurance agency, a real-estate operations manager, an interior design studio, a law office, an import company — alongside two roles that remain in accounting, still a real vertical), and the word "pilot" was removed from every place it appeared on the page (the trust-section card and one FAQ answer — a full

codebase search turned up no other occurrences). Gmail was dropped from every integration mention (Centro integrates with WhatsApp and Google Drive only); the Gmail icon, its orbit-animation entry, and one Gmail-referencing chip were replaced with non-Gmail equivalents.

The contact form (shared by the inline section and the timed popup) now requires an email address, and its "business name" field was replaced with an optional "business type" field with multi-industry example placeholder text, matching the same repositioning.

### **25.3 Real Bugs Found and Fixed**

Five genuine bugs were found through live device-oriented testing during this work, none of them part of the original request:

### 25.3.1 Hero headline invisible on mobile

The gradient-clipped "Centro" word in the Hero headline rendered as blank space on mobile. Root cause: it sat in a nested `<span dir="ltr" class="inline-block">` inside a parent using `bg-clip-text text-transparent` — `background` is not an inherited CSS property, and the nested `inline-block` created an independent rendering box that, on WebKit mobile specifically, did not receive the parent's clipped gradient. Fixed by removing the `inline-block`.

### 25.3.2 Document card rendering behind its own blur layer

The animated document card used `backdrop-blur-md`, which creates a compositing layer; on mobile Safari, the card's own transform-animated children (the flying PDF icon in `DocumentFlowAnimation`) rendered behind that layer instead of above it, appearing to vanish mid-animation. Fixed by making the blur desktop-only — it was barely visible on a card this small and opaque, so removing it on mobile has no real visual cost.

### 25.3.3 Missing-document chips overlapping on mobile

In `ProblemSection.tsx`, six scroll-animated chips are positioned with scroll-linked `x/y` coordinates scaled by a responsive `spread` multiplier. The multiplier shrank each chip's *position* on mobile, but never its rendered *size* (the text is `whitespace-nowrap`), so six full-width chips ended up crammed into a shrunk radius and overlapped. Fixed with a mobile-only static `flex-wrap` layout that reflows to the real viewport width; the desktop scroll-linked chaos-to-order animation is unchanged.

### 25.3.4 How It Works swipe carousel breaking under RTL

The new mobile swipe carousel (see 25.5) initially broke under the page's real `dir="rtl"`: a flex child anchors to the right edge of its container by default in RTL, so translating the track leftward to reveal a later step carried the whole track off-screen instead of sliding the next card into view. This is the exact same defect already documented and fixed in `TestimonialsSection.tsx`'s marquee. Fixed with the identical pattern — `dir="ltr"` on the track (so the transform math has a predictable anchor), `dir="rtl"` re-applied per card so the Hebrew content inside still reads correctly.

### 25.3.5 Orbiting integration icons — geometry, not just stacking

The first attempted fix for the orbiting integration icons (WhatsApp, Drive, PDF, Excel, AI) disappearing behind the document card on mobile raised their stacking order above the card. This masked the real problem rather than fixing it: the icons' orbit radius was still geometrically smaller than the card's own footprint at the mobile scale, so raising them above the card only changed *where* the overlap was visible — from behind the card to across its face. `HeroVisual.tsx` now measures the actual rendered card and container size at runtime (`ResizeObserver`, the same approach already used by `TestimonialsSection`'s marquee) and derives a real per-icon radius clamp: a floor

that clears the card's true worst-case corner distance with a real gap, and a ceiling that guarantees the icon never leaves the viewport. Icon size on mobile is scaled down separately from (smaller than) icon position, since each icon's own half-size otherwise eats into the available clearance budget twice over. On the narrowest phones (~360–390px) the two bounds can cross — the card's corner-to-corner distance genuinely does not fit the visible width at any radius — and the clamp resolves toward the visibility ceiling, so icons stay fully on-screen at all times even where a full gap isn't geometrically possible. Verified clear of the card with a real gap at 414px and wider; verified always fully within the viewport (zero clipping) at every tested width down to 360px. Desktop and tablet are unaffected — the new logic only engages below the existing `sm` breakpoint.



## 25.4 Popup Timing and Behavior

The timed demo-request popup ( `DemoRequestModal.tsx` ) opens automatically once per browser session, only on the landing page (it is mounted exclusively from `src/app/page.tsx` , never the shared app layout). Its trigger delay was changed twice during this work — 45s to 15s, then 15s to 35s — with only the delay constant changed each time; the trigger logic, the one-time-per-session guard (via `sessionStorage` ), and the defer-and-retry behavior when another overlay is open were never modified. A "does not appear" report was investigated directly rather than assumed fixed: a fresh browser context with no prior storage reliably showed the popup at the configured delay, and the specific edge case of the mobile menu being open at the trigger mark was also tested — the popup correctly deferred and then opened shortly after the menu closed. No code defect was found; the most likely explanation is a test browser tab retaining `sessionStorage` from earlier testing. The popup also gained a WhatsApp row beneath the form ("want to talk to us now?" / "here" button), linking to the same number and using the same official-style WhatsApp badge as the floating button.

## 25.5 Navigation and Interaction Fixes

- **Floating buttons** — the WhatsApp and accessibility buttons were restacked to the same mobile corner (bottom-left), identical 48×48px size, a real WhatsApp brand badge (green circle, white phone/speech-bubble glyph) replacing the previous brand-neutral icon for this specific "chat with us" use case.
- **Header** — a small "Log in" link was added (desktop and mobile); the mobile menu was rebuilt with a curated list rather than reusing the desktop nav array, removing entries that read as duplicates once "Watch how it works" and "Contact us" were added, and giving every item — including the FAQ link, previously styled differently from its neighbors — identical bordered-pill styling for visual consistency.
- **Hero and mobile-menu demo buttons** — both now smooth-scroll to `#demo` (the interactive demo section) instead of jumping instantly to `#final-cta` ; the mobile menu's primary CTA additionally closes the menu on click and opens no modal and no new page.
- **How It Works** — desktop keeps its original scroll-driven sticky panel unchanged, extracted verbatim into its own component. Mobile gets a new swipeable carousel instead (drag gesture, tap-to-navigate step numbers, pagination dots, no vertical scroll-locking — the page continues scrolling normally immediately after the section), since the scroll-jacked version required many small, unreliable scroll gestures on touch devices.

## 25.6 Testing Performed

Every fix in this chapter was verified against the running application, not inferred from the diff:

- `tsc --noEmit`, ESLint, and the full Vitest suite (138 tests across 18 files) clean after each of the three commits.
- Three full production builds ( `next build` ), one per commit, all clean.
- Live Playwright verification against a running dev server: real click/drag/tap interactions (not just screenshots) for the Hero and mobile-menu demo buttons, the popup trigger and its overlay-deferral edge case, the mobile How It Works carousel (tap navigation and an actual drag gesture), and DOM/computed-style assertions for the floating-button alignment and size, the mobile nav item styling, and the orbit-icon clearance geometry — measured, not eyeballed, across five real mobile widths (360, 375, 390, 414, 430px) plus desktop.
- A final pass re-ran the key assertions against the live production domain ( `www.centro-ai.co.il` ) after each deployment, confirming the built and deployed bundle matched what was tested locally.

## 25.7 Files Touched

| File                                                                              | Change                                                                                     |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <code>src/components/landing/TrialCta.tsx</code>                                  | new — shared trial-CTA component, default and inverse (white-on-color) variants            |
| <code>Hero.tsx</code>                                                             | headline fix, positioning line, trial CTA, demo-button smooth scroll to <code>#demo</code> |
| <code>HeroVisual.tsx</code>                                                       | mobile backdrop-blur removal; runtime orbit-radius measurement and clamp                   |
| <code>OrbitingIntegrations.tsx</code>                                             | per-icon radius clamp, separate mobile icon-size scale                                     |
| <code>ProblemSection.tsx</code>                                                   | mobile flex-wrap chip layout; Gmail chip replaced                                          |
| <code>HowItWorksSection.tsx</code>                                                | mobile swipe carousel (new), desktop extracted unchanged                                   |
| <code>FloatingWhatsAppButton.tsx</code> ,<br><code>AccessibilityWidget.tsx</code> | mobile restack, matched sizing, panel-offset fix                                           |
| <code>icons/IntegrationIcons.tsx</code>                                           | official-style WhatsApp badge added; Gmail glyph removed                                   |
| <code>DemoRequestModal.tsx</code>                                                 | trigger delay (45s → 15s → 35s); WhatsApp row added                                        |

|                                                                                                                                 |                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <code>Header.tsx</code>                                                                                                         | login link; mobile menu rebuilt (dedupe, contact link, consistent styling, demo-button behavior) |
| <code>TrustSection.tsx</code> , <code>FinalCTASection.tsx</code> ,<br><code>InteractiveDemoSection.tsx</code>                   | trial CTA placements 2–4; pilot language removed                                                 |
| <code>TestimonialsSection.tsx</code>                                                                                            | role diversification, neutral intro copy                                                         |
| <code>FAQSection.tsx</code> , <code>Footer.tsx</code> , <code>ContactForm.tsx</code> ,<br><code>api/contact/route.ts</code>     | copy rewrite, required email, business-type field                                                |
| <code>DashboardPreviewSection.tsx</code> ,<br><code>InteractiveDemoSection.tsx</code> ,<br><code>TestimonialsSection.tsx</code> | mobile-only backdrop-blur removal for scroll performance                                         |
| <code>src/app/layout.tsx</code>                                                                                                 | SEO description: Gmail and accounting-specific wording removed                                   |

## 25.8 Completion Status

All items from all three rounds of landing-page work are implemented, live-tested, and deployed to production. No known open issues remain on the landing page as of this writing. One pre-existing, unrelated defect was noted but intentionally left untouched as out of scope for this chapter's work: this report PDF's own pages 73–84 (see 25.9).

## 25.9 Note on This Report's Own Pages 73–84

While updating this document, pages 73 through 84 of the previously-existing PDF were found to contain a browser "file not found" error page repeated twelve times, rather than real content — evidently a defect from whatever process generated an earlier version of this file. Per this chapter's own scope (documenting landing-page work only), those pages were left exactly as found rather than silently altered; this note exists so the defect is recorded rather than rediscovered as a surprise. Fixing it would require regenerating those specific pages from their original source, which is outside this update.

---

*Addendum generated from the actual repository state — three real commits, a full tsc/lint/test/build pass on each, and Playwright-driven interaction testing against both a local dev server and the live production domain, across five real mobile viewport widths and desktop.*

## 26. Owner Dashboard — Internal Platform Control Center

---

|                         |                                                                                                                                                                                                                                                                                                                  |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope</b>            | New internal-only /owner surface plus the additive schema, session, and integration changes it required. No customer-facing route, page, or existing behavior was changed except two currently-silent failure paths that now also write an audit event, and requireSession()'s addition of one suspension check. |
| <b>Commits</b>          | 11 across Phases 0-5 (738cfa6, 2b96a5f, 4b7ba52, 5b1a510, b048cc4, 7e83c6e, 107e721, aba06ca, 969255b, plus this documentation update)                                                                                                                                                                           |
| <b>Database changes</b> | 3 new tables (platform_owners, platform_owner_sessions, platform_owner_audit_log), 1 new table (job_runs), 1 new column (organizations.suspendedAt) — migrations 0025-0027, all additive                                                                                                                         |
| <b>Verification</b>     | tsc --noEmit, ESLint, the full Vitest suite, and a production build clean after every phase; every phase additionally live-tested with Playwright against a running dev server, including a real end-to-end lockout/reactivate/control test for Phase 5                                                          |

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This chapter documents a new, internal-only “Owner Dashboard” — a control center for the person operating the Centro business itself, not any customer organization. It did not exist before this work: the application had no admin, owner, or platform-level role of any kind, and every existing page was scoped to a single organization's own session. The dashboard was scoped and built in five phases, each independently verified and shipped.

### 26.1 Why a Separate Authentication System

The platform owner is not a tenant, and the existing Session type's organizationId field is non-nullable — getSession() performs an inner join through sessions -> users -> organizations, and every session is therefore permanently bound to exactly one organization. Folding the owner into that model would have meant either inventing a dummy internal organization (which would then need filtering out of every cross-organization query the dashboard writes) or making organizationId nullable across the application's single hottest read path. Both were rejected.

Instead, the Owner Dashboard has its own minimal, parallel authentication system, deliberately isolated from the customer-facing one:

- platform\_owners — id, email, passwordHash, createdAt. No organizationId.
- platform\_owner\_sessions — the same recipe as the existing sessions table (an opaque random id stored server-side, the cookie holds only the id) with the organization dimension removed.
- platform\_owner\_audit\_log — mirrors the existing audit\_logs table's insert-only design, without its organizationId foreign key (which is NOT NULL there), and additionally carries a severity column (info / warning / critical) from day one so a future alerting mechanism needs no schema change.

A dedicated cookie (centro\_owner\_session, path-scoped to /owner) keeps an owner session and a customer session fully independent — they can coexist in the same browser without any risk of one escalating into the other. The guard function, requireOwnerSession() in src/lib/auth/ownerSession.ts, mirrors the existing requireSession() exactly, including this codebase's established convention of checking auth per-route rather than via Next.js middleware (there is no middleware.ts anywhere in this project, by deliberate prior decision). Every page, layout, and Server Action under src/app/owner/\*\* independently calls this guard — a layout only protects the pages it wraps, not the Server Actions those pages invoke, so each action file re-checks it, matching the existing convention already used for the customer-facing app's own protected actions.

Provisioning is CLI-only — `scripts/create-owner.ts` and `scripts/reset-owner-password.ts`, mirroring the existing `scripts/create-organization.ts` — with no self-registration route of any kind, ever. Login attempts are rate-limited using the existing `src/lib/auth/rateLimiter.ts` unmodified, and passwords are hashed with the existing script-based `src/lib/auth/password.ts` unmodified.

## 26.2 Phase 0 — Auth Foundation

Delivered the schema above, the `requireOwnerSession()` guard, the `/owner/login` page (a Server Action, not a Route Handler, so it keeps Next.js's automatic same-origin Origin-header protection), a guarded owner shell layout, and the CLI provisioning scripts. Verified end to end with a real login/logout cycle, an invalid-credentials rejection, and confirmation that no customer-facing cookie or route was affected.

A real implementation detail worth recording: the first attempt at visually distinguishing the owner shell from the customer app tried to override the shared design tokens (`--color-background`, `--color-text-primary`, etc.) inside a scoped CSS class, on the theory that `Card` / `KpiCard` / `Table` / `Badge` / `EmptyState` / `PageHeader` could then be reused completely unmodified in a dark palette. This does not work — confirmed directly, by overriding the custom property at runtime in a live browser and observing zero effect. Tailwind v4's `@theme` inline directive resolves every one of those utility classes to a literal color value at build time rather than emitting a `var()` reference, so a later CSS override has nothing to act on. The shared components are reused exactly as designed, in their normal light styling; visual distinction from the customer app comes instead from a small hand-written dark header chrome (`.centro-owner-header` in `globals.css`) that does not depend on the token system at all.

## 26.3 Phase 1 — Organization Directory and Global Search

New cross-organization queries in `src/lib/data/owner/organizations.ts` — the first functions in the entire data layer that intentionally drop the `organizationId` scoping every other query enforces. `listOrganizations(query)` generalizes the existing `searchClients()` ILIKE pattern across organizations and users; `getOrganizationOverview(id)` returns integration status, onboarding state, and usage counts for one organization. A nav-level search box in the owner shell's header searches organization name, email, and phone, and jumps straight to the matching organization's detail page. The detail page's activity tab reuses the existing `listAuditLog()` function completely unchanged.

## 26.4 Phase 2 — Executive Home Dashboard, Onboarding Funnel, Activity Feed

`/owner` is the default landing page: an 11-tile KPI grid (total and active organizations, new today/this month, open/completed/failed collection requests, documents processed today, AI messages today, WhatsApp messages today, Drive uploads today), a six-stage onboarding funnel chart, and a merged, reverse-chronological activity feed. “Active organization” is defined as any audit-log event — not only logins — in a trailing 30-day window, since a 30-day session lifetime and one shared login per organization make login-only tracking understate real usage. “Failed” collection requests are those in the escalated status; cancelled is tracked separately and not counted as a failure.

The activity feed merges the existing `audit_logs` table (generalized here to have no `organizationId` filter, unlike the customer-facing `/audit` page's own query) with the new `platform_owner_audit_log`. “Live” is implemented as always-current data on page load plus a lightweight client-side polling refresh every 45 seconds (a small `OwnerAutoRefresh` client component calling Next.js's `router.refresh()`) — not a push-based WebSocket or Server-Sent-Events stream, since this application has no real-time transport anywhere and building one was judged out of proportion to this feature. A genuine push-based upgrade remains a natural later addition on the

same event sources, not a rewrite.

The onboarding funnel (registered -> completed onboarding -> connected WhatsApp -> connected Google Drive -> created first collection request -> completed first collection request) is rendered with recharts, a new dependency — the first charting library this codebase has ever used — inside one small client-side wrapper component; the underlying counts are still fetched entirely server-side, consistent with the rest of the application.

## 26.5 Phase 3 — System Health and Alerting Foundation

A new `job_runs` table is written on every invocation of the existing `POST /api/cron/tick` endpoint, on both success and failure — previously that endpoint's {evaluated, reminded, delivered} counts only ever reached whichever external scheduler called it, once, and were otherwise lost; there was no way to answer “did last night's scheduled tick actually run, and did it fail” after the fact.

Two integration failure paths that were previously `console.error`-only are now also written to the existing per-organization `audit_logs` table, tagged with a severity value inside its metadata column: WhatsApp outbound send failures (`src/lib/conversationOrchestration.ts`'s `sendViaWhatsApp`) and Google Drive token-refresh failures (`src/lib/googleAuth/driveTokens.ts`'s `getValidAccessToken`). Both are wrapped so that a failure while recording the failure can never mask the original error or change either function's existing non-fatal behavior — confirmed by the full pre-existing test suite passing unchanged.

`computeHealthStatus()` (`src/lib/owner/health.ts`) is a small, pure function — no database access — over three windowed (trailing 24-hour) signals: failed job runs, warning-severity events, and critical-severity events. It is unit-tested in isolation (`health.test.ts`) and always returns its concrete reasons in Hebrew rather than a bare color, so a future automated notifier can call the same function and diff it against the last-known result instead of needing new detection logic. It is surfaced as a green/yellow/red badge next to the home dashboard's title.

## 26.6 Phase 4 — AI Usage and Cost Tracking

The internal AI Copilot chat (built in an earlier milestone) already captured token usage on every assistant turn, in `aiMessages.metadata`, but nothing in the application ever read it. A new static, manually-maintained pricing table (`src/lib/owner/aiPricing.ts`, \$ per million tokens by provider and model — no provider API or SDK returns price, only token counts, so this table has to be kept in sync by hand as real pricing changes) is combined with a new aggregation query (`src/lib/data/owner/costs.ts`) grouping usage by provider and model, for both “today” and “last 30 days.” A model missing from the pricing table still reports its real token counts, with its cost shown as “unavailable” rather than silently computed as \$0.

A new `/owner/costs` page shows the full breakdown; the home dashboard additionally gets an “AI cost today” tile. That tile uses a new `OwnerCurrencyKpiCard` component rather than the existing `KpiCard`, because `KpiCard`'s count-up animation always rounds its displayed value to the nearest integer — correct for a plain count, but it would collapse a real dollar estimate such as \$0.0075 down to a meaningless “\$0”. This was verified end to end against a real injected usage record (1,000 input and 500 output tokens on `gpt-4o`), which computed to exactly \$0.0075 on both the costs page and the home tile.

## 26.7 Phase 5 — Organization Suspend / Reactivate

The final phase, and the only one that can affect a real customer if built incorrectly, so it was sequenced last and tested the most heavily. A single nullable `organizations.suspendedAt` column is read as part of `getSession()`'s existing `organizations` join — costing zero additional database queries on the application's

hottest read path. `getSession()` itself was deliberately left returning a normal `Session` object for a suspended organization, unchanged: several existing callers (the login page, `redirectAfterAuth()`) depend on its null/non-null return contract as-is. The suspension check lives solely inside `requireSession()`, the one guard every protected customer-facing page and action already calls, which now redirects to a new `/suspended` page instead of returning.

`/suspended` reads `getSession()` directly — never `requireSession()` — specifically so it can never redirect into itself. It shows a clear, non-technical explanation, a WhatsApp support link (reusing the same `WHATSAPP_NUMBER` constant the rest of the application already uses), and a logout button.

Suspend and reactivate are owner-only Server Actions on an organization's detail page (`src/app/owner/(dashboard)/organizations/[id]/actions.ts`); suspending sits behind the same `ConfirmDialog` destructive-action pattern already used elsewhere in the application. Both actions are logged to `platform_owner_audit_log`.

This phase was verified with a real, scripted end-to-end scenario rather than inference from the diff: a real customer organization logs in normally; the owner suspends it; the customer's already-open session is confirmed to be redirected to `/suspended` on its very next navigation, with no further access to any protected page; a fresh login attempt with fully correct credentials is confirmed to still land on `/suspended`, not to bypass the suspension; the owner reactivates the organization and normal access is confirmed restored; and a second, control organization that is never touched is confirmed to be completely unaffected throughout. All 143 pre-existing automated tests continued to pass unchanged after this phase, including through the `session.ts` modification.

## 26.8 Hebrew, RTL, and Localization

Every owner-facing string is Hebrew by default and sourced from a small message catalog (`src/lib/owner/i18n/`) rather than hardcoded inline, which is the one place in this codebase that pattern is used — the customer-facing application hardcodes Hebrew directly in JSX throughout, by established convention, since it has never needed more than one language. The owner surface is structured differently on purpose: a flat key -> Hebrew-string catalog plus a small `t(key, params)` helper, hardcoded to Hebrew for this release, but shaped so that adding a second language later means adding a sibling catalog file and a resolution mechanism, not rewriting any component. The entire surface renders `dir="rtl"`, matching the rest of the application; the one chart (the onboarding funnel) is deliberately rendered `dir="ltr"` internally, since the charting library has no right-to-left layout mode of its own, while its Hebrew category labels render correctly regardless.

## 26.9 Routes and Files Reference

|                                                                                      |                                                                                                                                                            |
|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><code>/owner/login</code>, <code>/owner</code></b>                                | Owner login page and the default home dashboard                                                                                                            |
| <b><code>/owner/organizations</code>,<br/><code>/owner/organizations/[id]</code></b> | Organization directory, search, and detail (info, integrations, usage, activity, suspend/reactivate)                                                       |
| <b><code>/owner/costs</code></b>                                                     | AI usage and estimated-cost breakdown (today and last 30 days)                                                                                             |
| <b><code>/suspended</code></b>                                                       | Customer-facing landing page for a suspended organization's session                                                                                        |
| <b><code>src/lib/auth/ownerSession.ts</code></b>                                     | Owner session guard — <code>createOwnerSession</code> / <code>getOwnerSession</code> / <code>requireOwnerSession</code> / <code>destroyOwnerSession</code> |
| <b><code>src/lib/owner/audit.ts</code></b>                                           | <code>recordOwnerAuditEvent</code> — the one write path onto <code>platform_owner_audit_log</code>                                                         |

|                                                         |                                                                                                         |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>src/lib/data/owner/*.ts</b>                          | Cross-organization data layer: organizations, metrics, funnel, activity, health, costs                  |
| <b>src/lib/owner/health.ts, aiPricing.ts</b>            | Pure computeHealthStatus() logic and the static AI pricing table                                        |
| <b>src/db/schema.ts</b>                                 | platform_owners, platform_owner_sessions, platform_owner_audit_log, job_runs, organizations.suspendedAt |
| <b>scripts/create-owner.ts, reset-owner-password.ts</b> | CLI-only owner provisioning and password recovery                                                       |

## 26.10 Testing Performed

Every phase was verified with the same discipline used throughout this report: tsc --noEmit, ESLint, the full pre-existing Vitest suite, and a production build (next build), all clean, before that phase was considered complete — with the pre-existing test count growing from 138 to 143 (five new tests for computeHealthStatus()) and every single pre-existing test continuing to pass unchanged through all five phases, including the two phases (Phase 3 and Phase 5) that modified shared, already-live files (conversationOrchestration.ts, driveTokens.ts, session.ts).

Beyond the automated suite, every phase was additionally live-tested with Playwright against a real running dev server rather than inferred from the diff: real login/logout cycles for both the owner and customer auth systems; real search and navigation through the organization directory; injected job-run and audit-log records to confirm the health badge escalates from green through yellow to red with the correct Hebrew reasons attached; an injected AI usage record to confirm the cost calculation is exactly correct, not merely plausible; and the full suspend/reactivate/control scenario described in Section 26.7 above.

One environment lesson surfaced and is recorded here since it cost real debugging time during this work: several long-orphaned Node processes from earlier, unrelated sessions were found still running in the background, silently holding the local PGlite development database open. PGlite does not support two separate OS processes holding the same data directory open at once (an existing, documented limitation of this project's local dev setup) — the symptom was cross-organization counts that were internally consistent within any one process but disagreed between the running dev server and a freshly-run script. Once every orphaned process was identified and terminated and the local database was reset from a clean migration, every count matched exactly, including under concurrent Promise.all-issued queries. No application code was at fault; this is noted here as a real operational finding, not a defect in the feature.

## 26.11 Deferred Scope

Per the approved plan, three items were explicitly scoped out of this build and left for a later, separate initiative: manual infrastructure/WhatsApp cost entry and an approximate “estimated profit” figure; real alert delivery (email or WhatsApp) built on top of computeHealthStatus() and job\_runs; and a real billing/subscription system — without which “paying users,” MRR, churn rate, and true revenue have no backing data anywhere in the application and were correctly omitted from this build rather than faked.

---

Addendum generated from the actual repository state — eleven real commits, a full tsc/lint/test/build pass after every one of them, and Playwright-driven interaction testing against a real running dev server, including a full real-session suspend/reactivate/control lockout test.



## Addendum to Chapter 26 — Production Deployment & Verification

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|                            |                                                                                                                                                                                                                                  |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Production URL</b>      | <code>https://www.centro-ai.co.il</code> (Vercel project <code>centro-ai</code> )                                                                                                                                                |
| <b>Deployed commit</b>     | 4363bda (Add Chapter 26 to Implementation Report: Owner Dashboard, Phases 0-5) — the tip of <code>main</code> at the time of this verification, confirmed by inspecting the live Vercel deployment directly                      |
| <b>Verification method</b> | Safe, read-only and side-effect-free checks only, from outside the application, plus inspection of Vercel's own project configuration and runtime logs. No direct production database access was used or available at any point. |

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### Deployment mechanism

No manual deployment step was required. The Vercel project is connected to this repository's GitHub remote for automatic deployment on every push to `main`; every commit pushed across Phases 0-5 and the Chapter 26 documentation update was already live in production within minutes of being pushed, confirmed by inspecting the Vercel deployment history directly (`vercel ls / vercel inspect`) and matching the deployed commit hash to the local git HEAD.

### Database migrations — how they were verified without direct access

The production `DATABASE_URL` is configured in Vercel as a "Sensitive" environment variable — a real Vercel platform feature meaning its value cannot be read back by anyone, including the project owner, through the dashboard or CLI once set. This is a genuine security boundary, not a permissions gap, and it was respected rather than worked around: no attempt was made to obtain or reconstruct the production connection string.

Migration application was instead confirmed live, functionally: submitting a deliberately-invalid owner login attempt against the real production site returned the correct Hebrew "invalid credentials" error rather than a server error, which is only possible if the `platform_owners` table already exists and is queryable — an unrecognized email never causes a database write, so this check has no side effects. Vercel's runtime logs for the verification window were also inspected directly and contained zero error-level entries. Server logs additionally showed real prior requests to `/owner/organizations/[id]` with genuine organization identifiers, indicating the dashboard's deeper functionality (which depends on the same migrations) was already being used successfully in production before this verification pass began.

### Owner account provisioning

The real owner account was provisioned by the account holder directly, running the existing `scripts/create-owner.ts` CLI script themselves against the production database, with the production `DATABASE_URL` set only in their own local shell for that one command. The email and password were never entered into, generated by, displayed to, or transmitted through the assistant at any point, and nothing resembling a credential was written to this repository, a commit, or a log — preserving the design decision recorded in Chapter 26 that the platform-owner account has no public or assistant-mediated provisioning path of any kind.

### Production verification performed

- `/owner`, `/owner/organizations`, `/owner/costs` (unauthenticated): each correctly redirects (HTTP 307) to `/owner/login` rather than exposing any content.
- `/owner/login`: renders correctly, `dir="rtl"`, Hebrew heading, on both desktop and mobile (390px) viewports.

- A deliberately-invalid owner login attempt: rejected with the correct Hebrew error message, confirming the platform\_owners table is live (see above) with no side effects.
- /suspended, /dashboard (unauthenticated): both correctly redirect to /login.
- /login (customer-facing) and /api/health: unaffected, rendering/responding normally.
- Vercel runtime logs for the verification window: zero error-level entries.

## Scope

This addendum documents verification only. No application code, schema, or configuration was changed as part of this production-readiness pass — Phases 0-5 were already complete and already deployed automatically as a side effect of the commits documented in Chapter 26. Phase 6 (manual cost/profit) and Phase 7 (billing/subscriptions) remain explicitly out of scope and were not started.

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Addendum generated from the actual production state — live HTTP verification against [www.centro-ai.co.il](http://www.centro-ai.co.il) and direct inspection of the connected Vercel project and its runtime logs, with no direct production database access at any point.

## Chapter — Product Evolution M9

Recurring vs. On-Demand Collection: Automatic Scheduling, Trustworthy Classification, and Mandatory, Recoverable Integrations

### Why this chapter exists

An audit of the existing recurring-collection workflow (see ARCHITECTURE.md, Chapter 9) surfaced a specific, load-bearing gap: Centro stored a "collection day of month" but nothing ever auto-created the next Collection Request from it. Every cycle was opened by an employee, by hand, every single time. This chapter records the implementation that closed that gap end to end, along with several related trust and reliability issues discovered in the same audit.

### From a permanent lock to a per-item choice

Before this milestone, "recurring" vs. "one-time" was a permanent, organization-wide decision made once at onboarding (organizations.workflowType). It is now a property of each individual Service (services.collectionMode: "recurring" or "on\_demand") — one organization can run both kinds side by side. Onboarding's Collection Style step now offers a third choice, "Both," alongside the renamed "Recurring Collection" and "On-Demand Collection" concepts (Hebrew: Isuf Machzori / Isuf Lefi Tzorech). The navigation now always shows both areas once onboarding completes, and a new Service or Template can be created at any time thereafter.

### The automatic recurring engine

A new module, src/lib/recurringScheduler.ts, is the engine this milestone was built around. Each Recurring Service gets a configurable frequency (services.collectionFrequencyIntervalMonths — "every X months," covering monthly, every-2-months, quarterly, every-6-months, annual, and any custom interval with a single field). Each client assignment tracks its own next-due date (client\_services.nextCollectionRunAt). A new cron pass, run every tick alongside the existing reminder and inactivity passes, finds every pairing that has come due, snapshots the Service's document requirements, opens a new Collection Request, and sends the initial request through the exact same delivery path a manual send already used — then advances the schedule from where it was supposed to fire, not from "now," so a late tick never drifts the cadence forward.

The guarantee that an On-Demand Collection can never auto-repeat is enforced server-side, not by UI convention: the query that finds due work filters on services.collectionMode = "recurring" as a hard database WHERE clause. There is no code path by which an On-Demand Service could be picked up — not a flag a future change could forget to check.

This was verified directly against a live database: a recurring pairing marked due had a new Collection Request created for it, its schedule correctly advanced to the next occurrence, a second immediate run created nothing further (no duplicate), and an on-demand pairing present in the same organization was never touched across the entire test.

### Frequency and reminders, modeled as separate concepts

Collection frequency (when the next cycle starts) and reminder rules (what happens inside an already-open cycle still waiting on documents) are now presented and configured as explicitly distinct settings, each with its own card in the Service configuration screen and in onboarding — closing a real source of confusion in the previous single combined "reminder rules" step.

## AI suggests, the user approves

Before this milestone, an automatic classification of an imported or manually created client (from a spreadsheet's business-type column, or inferred from context) was written and made immediately effective in the same step — the client was linked to that type's Service, and its document requirements applied, before any human had reviewed the guess. A new column, `clients.classificationConfirmedAt`, separates "Centro suggested this" from "a human approved this." An automatic classification is now visible in the UI as a clearly labeled suggestion but has no effect on what is requested from that client until a person confirms it — either by correcting it, or by approving it as-is. A new bulk "Confirm" action approves every pending suggestion for one business type in a single click, so this discipline does not become tedious for an office importing hundreds of clients at once. When a spreadsheet has no usable type information at all, Centro still never guesses; the existing manual-assignment screen gained a search filter and a select-all-filtered control so dividing a large list into groups by hand stays fast.

## WhatsApp and Google Drive: mandatory, verified, and recoverable

Both integrations were previously best-effort — onboarding could complete, and automation could appear "ready," without either one actually connected. Onboarding completion is now hard-blocked until a live check (`src/lib/integrationRequirements.ts`) confirms WhatsApp's identifiers are present and Google Drive's access token can actually be refreshed — not merely that a connection happened at some point in the past. The same check gates the moment a collection is actually started, surfacing a specific, plain-language explanation and a direct link back to Settings rather than a generic error. Settings itself gained a full reconnect/disconnect panel for both integrations, closing a real gap where neither could previously be managed after onboarding finished.

## Never losing a document to a temporary Drive outage

Previously, a document's temporarily held bytes were cleared the moment it was marked approved, before Centro knew whether the Google Drive upload had actually succeeded — a failure immediately after approval meant the real file was gone for good. Those bytes are now cleared only once the upload is confirmed successful. A failure is recorded and automatically retried on every subsequent cron tick, up to eight consecutive attempts, before being escalated as a critical, clearly logged event requiring manual attention — the document's real bytes remain safely held and recoverable throughout, never silently discarded.

## Disabling automation now actually disables it

The existing "Deactivate automation" toggle already behaved like a real on/off switch in the database, but the message-sending pipeline never actually checked it. It now does: deactivating stops every automatic send and every automatic cycle-creation attempt organization-wide, immediately, without touching any historical data. A narrower, per-Service pause control was added alongside it for pausing one specific recurring collection without affecting any other. Both controls carry a short, plain-language explanation of exactly what they do, directly in the interface.

## Verification performed

- Full registration through onboarding completion, live, including the third "Both" collection-mode option.
- The critical hard-block test: attempting to finish onboarding without connecting WhatsApp and Google Drive was correctly rejected with a specific, honest explanation, live, not merely reviewed in code.
- Classification suggestion and confirmation: an imported spreadsheet's explicit business-type values produced pending suggestions, not immediate authoritative assignments; confirming one group reduced the pending count while leaving other groups untouched.
- Industry-specific personalization: an office declaring itself a business consultant was correctly offered

business-consultant-relevant starting document types, never the accounting-specific ones, and vice versa for an accounting office.

- Pause and resume of a live recurring collection, and the organization-wide automation toggle, both confirmed live with the correct resulting state shown in the interface.
- A live-triggered blocked send (WhatsApp disconnected) correctly showed the required plain-language message and a working link back to Settings.
- The automatic scheduling engine itself, directly against a live database: a due cycle was created, the schedule correctly advanced, an immediate re-run created nothing further, and the on-demand collection in the same organization was never auto-created across the entire test.
- TypeScript compilation, ESLint, the full automated test suite (159 tests, including 10 new tests covering the scheduling date math), and a full production build all passed cleanly.

This work deliberately did not touch the existing one-time/on-demand user journey's own product decisions, and did not touch the separately paused Grow/payment/subscription plan.

## Chapter — Smart Profession-Aware Onboarding

Recurring, On-Demand, and Both: closing the gap between an accounting-first origin and a multi-profession product

### Why this chapter exists

The onboarding wizard was originally built accounting-firm-first. As Centro grew to serve other professions (insurance, law, real estate, mortgage advising, HR, finance, and free-text ‘other’ categories), the product owner asked for a full audit of the onboarding experience: does the selected profession genuinely control what the wizard shows, or does accounting-specific language and logic still leak through regardless of what a new office actually does? A read-first audit (three parallel investigations across the wizard’s step components, server actions, the business-type/template schema, and the AI-classification pipeline) found the product was already far more profession-aware than assumed — nine business categories already had distinct, curated starter document sets, and the five hardcoded accounting starter types were already gated to the accounting/tax-advisor categories only. This chapter records the six specific, narrower gaps that audit actually found, plus a dedicated final QA pass that caught what the code-only audit had missed.

### A classifier with no opinion about its own candidates

The client business-type classifier matches imported spreadsheet text against a dictionary of Israeli accounting terms (Limited Company, Authorized Dealer, Exempt Dealer, and similar) unconditionally, regardless of the organization’s declared profession. Before this change, an insurance organization was protected from a misclassification only by accident — its own candidate list, seeded per its declared category, never contained an accounting canonical key, so a dictionary hit for an accounting term always fell through to a plain, unexplained ‘unclassified.’ The classifier now reports this case explicitly as a conflict instead of silently discarding the signal, and the onboarding wizard’s classification-review step shows these by name: which client, what text matched, and which out-of-profession business type it resembles. Nothing is auto-applied from a conflict; it exists purely so the office can look.

### Templates without Excel or AI

The manual business-type/template creation flow already existed outside onboarding, at the dedicated Templates and Services pages — what onboarding itself never offered was a way to reach that same creation flow from inside the wizard without first uploading a spreadsheet. A persistent, secondary ‘create your own’ option was added to the import and classification-review steps, deliberately not a fork replacing the spreadsheet/AI path — an office can mix both in the same session. A manually-created entry is just another row in the same underlying table the AI-suggested entries populate, so every downstream step (required documents, reminders, summary) needed no changes to treat it identically.

### One bulk-assignment component, not two

Because a manually-created business type is indistinguishable, at the data level, from an

AI-suggested one, the classification-review step's existing bulk filter/select-all/assign interaction was extracted into its own shared component rather than building a second, parallel one for manually-created entries. Explicit 'leave unassigned' required no new mechanism — it is simply the default state the moment nothing is selected.

### **Excel rows get one more look before they become clients**

Column-mapping confirmation already existed as a review step, but only for how spreadsheet columns map to roles (name, phone, business type). The parsed rows themselves previously went straight into the clients table the moment mapping was resolved, whether resolved automatically or by hand. Both the automatic-detection path and the manual-mapping path now return a row-level preview instead of writing anything; a new confirmation action is the only path that actually performs the write, and it runs only after the office has seen every parsed row and could deselect any of them. This is the concrete form of 'the spreadsheet is context, never final truth' for row data, extending the same principle already established for classification suggestions.

### **The Connect step becomes a real checkpoint**

Product Evolution M9 (see the preceding chapter) made WhatsApp and Google Drive a hard-blocking requirement, verified live rather than trusted from a stored flag — but the only place that check actually ran was the wizard's final 'Go to Dashboard' action, which then bounced an unconnected office all the way back to the Connect step with an error banner. Nothing prevented clicking through five intermediate steps first with neither integration connected. The Connect step itself now computes the same readiness check and simply does not render a working Continue button until both integrations are verified, with a plain-language reason shown inline. No client-side polling was required: every path that changes connection state — the Google OAuth callback, the WhatsApp popup's completion handler, folder selection, disconnection — already forced a fresh render of this exact screen before this change, so the same server-computed readiness value the button now gates on was already correct on every render. The original hard-block check remains in place as defense in depth; this change closes the user-experience gap, not the only real enforcement.

### **Staying oriented: which profession is this, again?**

The declared profession was chosen once, at step three, and never referenced again by any later step — a real, if minor, way to lose track of what is being configured eight steps later. The wizard's shared header now shows a small profession badge (label and icon) from step four onward, once a profession has actually been chosen.

### **Final QA pass**

A dedicated end-to-end review followed — reading actual rendered screens rather than only source — before the project was closed, and it caught several things the original, code-only audit had missed precisely because they were not classification concerns: the Connect step's own help text still said the integrations could be connected later, directly contradicting the hard checkpoint just built; the collection-style step's own tooltip and the business-name field's placeholder both defaulted

to accounting examples regardless of the profession already chosen; the new manual-creation UI said 'template' while every surrounding screen already said 'business type' for the same entity; a disabled Continue button and the new profession badge had each been styled by hand instead of reusing the application's existing disabled-button and badge components; and the required-documents step's empty state was a plain paragraph, unlike its neighboring steps, and pointed to the wrong page name. Every one of these was a text or consistency correction, not new behavior, and all three onboarding flows were re-verified live afterward.

## Verification performed

- Recurring flow, live, for an insurance-category organization: profession badge visible from step four onward; Continue at the Connect step genuinely disabled with neither integration connected, and genuinely enabled once both were; a manually-created business type created inline and correctly appearing alongside AI-suggested ones; a spreadsheet import producing a row-level review screen before any write; an intentionally-included accounting-only term in the test file correctly surfaced as a conflict rather than silently misclassified or silently dropped; classification confirmation and the shared bulk-assignment panel both exercised; full completion through to the dashboard.
- On-Demand and Both flows, live, each carried through to the dashboard with no regression from the Connect-step change, which is shared code across all three flows.
- A dedicated check of a Recurring organization with zero business types configured, to confirm the corrected empty state and its corrected page reference.
- TypeScript compilation, ESLint, and the full automated test suite all passed cleanly after each round of changes, and again after the final QA pass; a full production build succeeded.

This work did not touch the Grow/payment/subscription plan, the Owner Dashboard, or conversion analytics, all of which remain separate, unstarted projects.